WELCOME TO SUGARCUBE™



Thank You for purchasing the SugarCube **SC1-Plus**.

Your experience of high-end audio is about to get a lot sweeter when all your vinyl will sound like mint.

These pages provide an overview of the set-up and control of the **SC1-Plus**.

For any usage questions, please contact:
support@sweetvinyl.com

SugarCube

Table of Contents

Section 1 Unpacking Your SugarCube ..... p 3

Section 2 Connecting SugarCube to Your Sound System ..... p 7

Section 3 Networking the SugarCube ..... p 16

Section 3.1 Using Ethernet or Wi-Fi with the SugarCube ..... p 17

Section 3.2 Accessing the SugarCube with a Browser ..... p 22

Section 3.3 Using the iOS app with the SugarCube ..... p 31

Section 3.4 Using the Android app with the SugarCube ..... p 43

Section 4 Controlling the SugarCube via a browser or app ... p 52

Section 5 Connecting a computer to the Plus via USB ..... p 64

Section 6 Technical Specs ..... p 87

Section 7 FAQ ..... p 91

Section 1

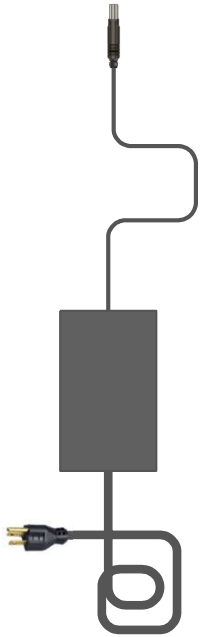
Unpacking Your SugarCube



1.1 **Unpacking** Your SugarCube



SugarCube SC1-Plus Unit



AC Power Adapter, 5V, 4A DC Output



Wifi USB Adapter

This adapter connects the SugarCube to your WiFi network or you can use a wired Internet connection if available.



USB Stick

USB-stick contains the full SugarCube Setup and User Guide.

Please download the **SugarCube App** from your mobile device's app store or use a **Browser** to control your **SugarCube** remotely.

1.2 SC1-Plus **Front-Panel Controls**



BYPASS

When pressed and the LED is lit, an internal relay has shunted the **LINE IN** directly to **LINE OUT** completely bypassing all internal circuitry.

CLICK REMOVE

When pressed and the LED is lit, SugarCube is processing audio to remove clicks and pops. The **BYPASS** LED will go out.

CLICK MON

When pressed and the LED is lit, the SugarCube suppresses the music so that you can better hear the clicks and pops. This helps set the **STRENGTH** level for particularly noisy LPs.

STRENGTH

Turn knob to control the strength of the click and pop detection and removal process. Almost all LPs will sound great at a setting of 4 or 5.

Note: we recommend that you do **NOT** adjust **STRENGTH** all the way to **10**, you will rarely need this much. Sometimes less is more.

1.3 SC1-Plus **Back-Panel**



DIGITAL OUTPUT
Toslink and SPDIF outputs at 192K/24bit.

DC POWER IN
Connect the supplied adapter to this port.

POWER ON/OFF
Note: when power is off, an internal relay bypasses **LINE IN** to **LINE OUT**.

USB COMPUTER PORT
Use this port to link your computer or music server to for enhanced ADC and DAC functionality. Detailed instructions later in guide.

WIRED LAN
If an Ethernet LAN cable is available, connect to this port instead of plugging in the WiFi adapter.

LINE OUTPUT
Outputs to your line stage (pre-amp). When the front panel **BYPASS** button is pressed, the audio from **LINE IN** passes directly to this **LINE OUT** through a mechanical audiophile-grade relay.

LINE INPUT
Accepts the output of your phono pre-amp. Please do NOT directly plug your turntable into this port unless it has a built-in phono pre-amp with a line-level output.

USB PORT FOR WIFI ADAPTER

Do either of the above when controlling the SC1-Plus on the SugarCube iOS or Android app or from your desktop.

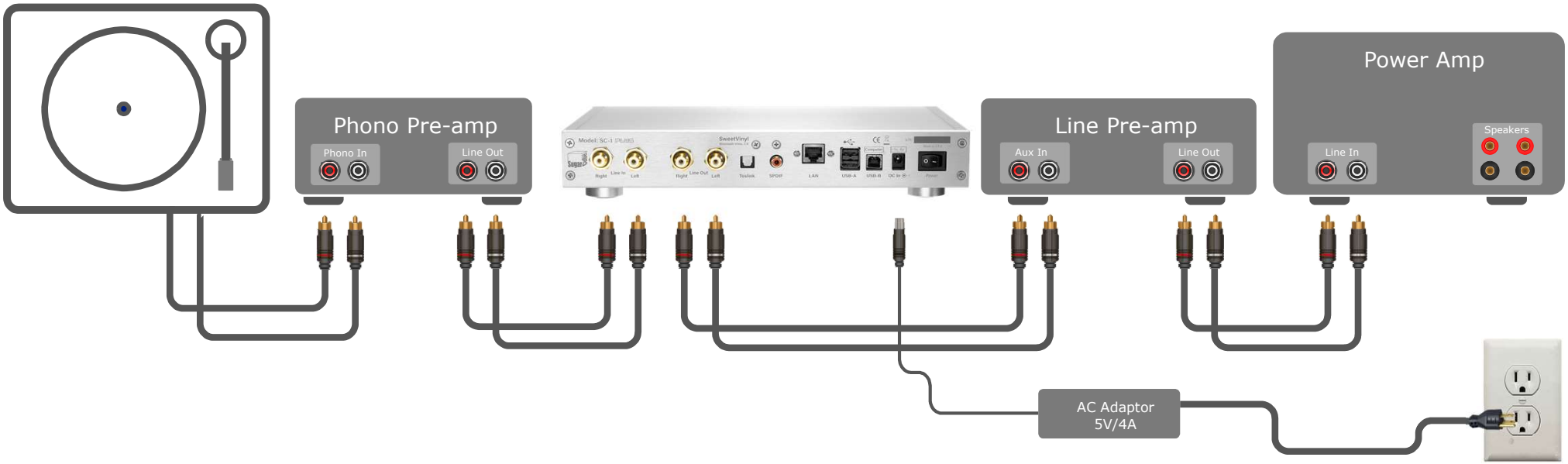
Section 2

Connecting SugarCube to Your Sound System

PLEASE BE SURE ALL EQUIPMENT IS TURNED OFF BEFORE ATTEMPTING INSTALLATION



2.1 Connecting Your SugarCube

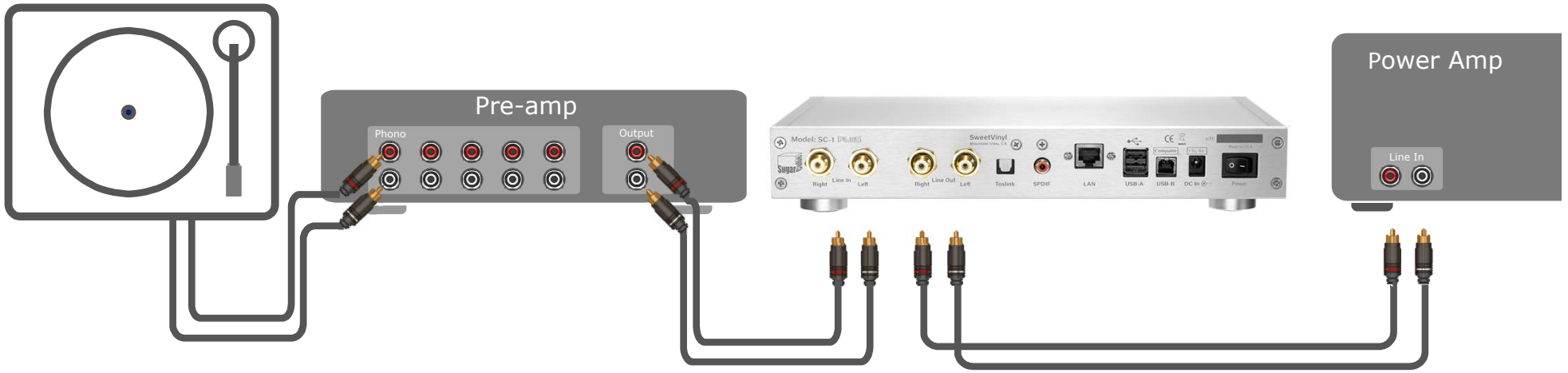


Connecting in series between a phono pre-amp and line-stage (pre-amp).

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your phono pre-amp to the **LINE IN** of the SC-1 Plus.
3. Connect the **LINE OUT** of the SC-1 Plus to an appropriate line-level input of your line-stage/pre-amp.

NOTE: DO NOT CONNECT THE **LINE OUT** OF THE **SUGARCUBE** TO THE PHONO INPUT ON YOUR EQUIPMENT.

2.2 **Connecting** Your SugarCube

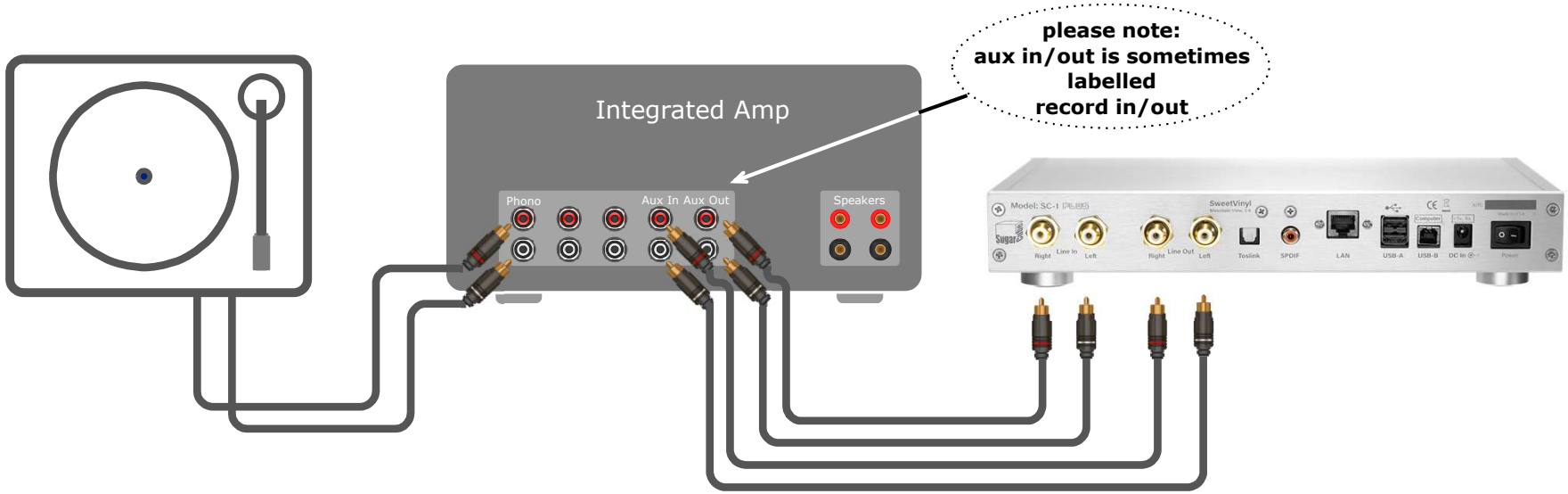


Connecting in series between a pre-amp, with phono input, and power amp

1. Connect the output of your pre-amp to the **LINE IN** of the SC1-Plus.
2. Connect the **LINE OUT** of the SC1-Plus to the input of your power amp.

2.3 **Connecting** Your SugarCube

Integrated Amp: using the 'tape loop'



Connecting to an integrated amplifier via Aux In/Out (or Rec In/Out).

1. Connect **AUX OUT** (or REC OUT) of your integrated amp (or processor) to the **LINE IN** of the SC1-Plus.
2. Connect **LINE OUT** of the SC1-Plus to **AUX IN** (or **REC IN**) of your integrated amp or processor.

NOTE: DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.4 **Connecting** Your SugarCube

Integrated Amp: connecting between the preamp and power amp

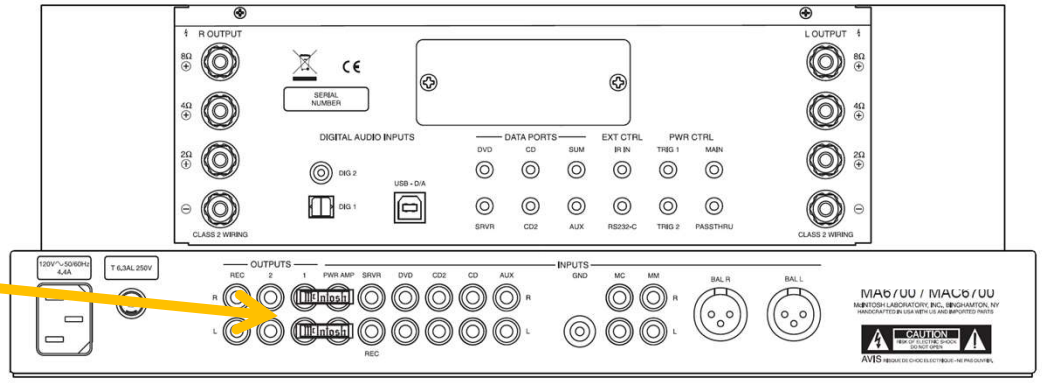
Step 1: TURN OFF THE INTEGRATED AMP

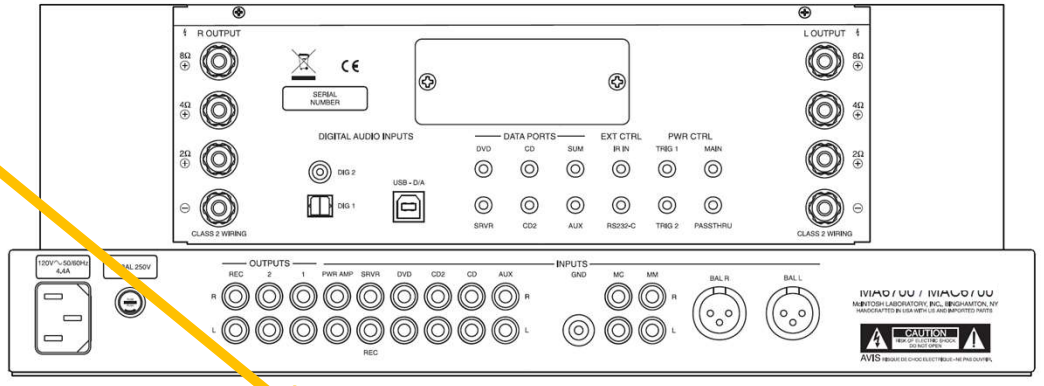
Step 2: Wait a minute or two

Step 3: locate the preamp-power amp jumpers

Step 4: DOUBLE-CHECK THAT THE AMP IS OFF

Step 5: Remove the jumpers





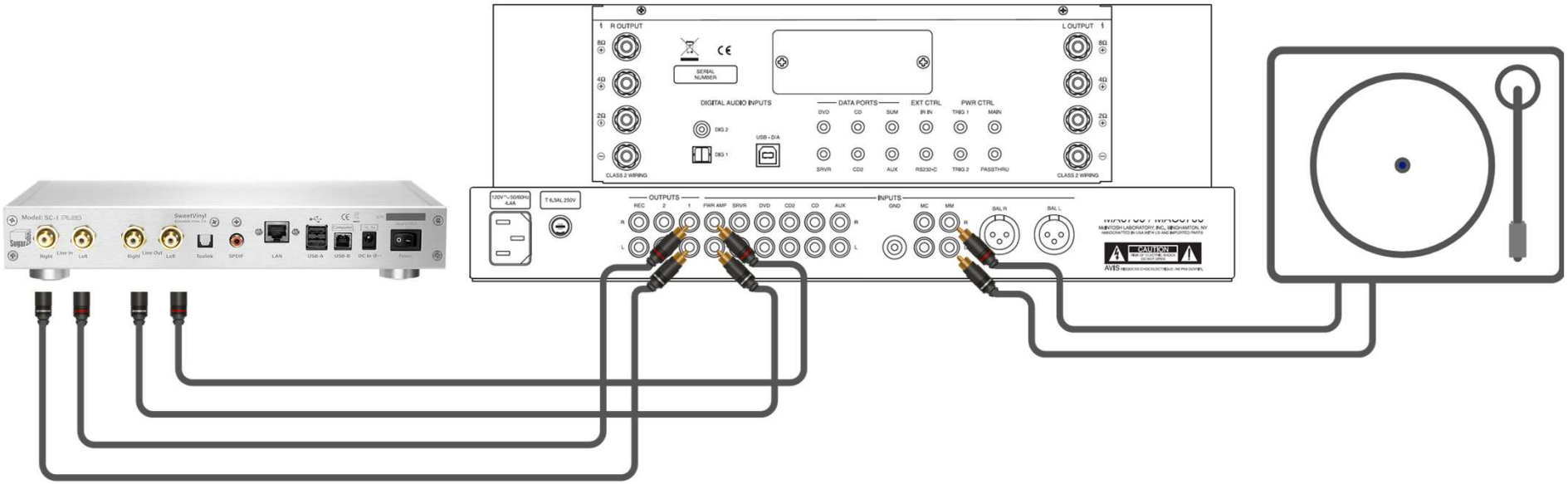


Step 6: Put the jumpers away in a safe place

Step 7: NEXT PAGE...

2.5 **Connecting** Your SugarCube

Integrated Amp: connecting between the preamp and power amp, cont.



Connecting to an integrated amplifier between the pre-amp and power-amp.

1. Connect **OUTPUT** (this is the pre-amp) of your integrated amp to the **LINE IN** of the SugarCube.
2. Connect **LINE OUT** of the SugarCube to **PWR AMP** (power amp input) of your integrated amp.
3. ALWAYS TURN ON THE SUGARCUBE FIRST THEN THE AMP AND TURN OFF THE AMP FIRST THEN THE SUGARCUBE.
4. If the integrated amp has a headphone output, it will likely have its own amplifier and may not be in the loop with the SugarCube. If that is the case, you will not hear the SugarCube in the headphones.
5. The SugarCube draws only about 11 watts, you can leave it on full time if you wish.
6. Not all integrated amps have a line out / power amp in or an Aux Out / In. If this is the case, you may need an external phono pre-amp.

2.6 **Connecting** Your SugarCube

Integrated Amp **WITHOUT** an AUX OUT/IN (or REC OUT/IN, etc.)



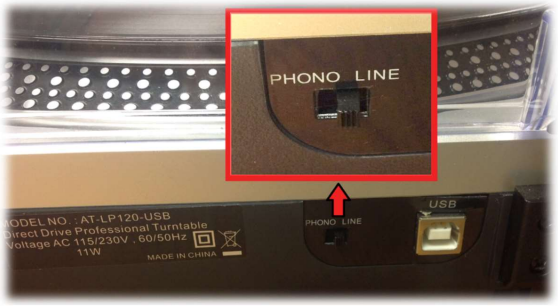
Many contemporary integrated amplifiers no longer have what was known as a ‘tape loop’ (back in the day of the cassette and reel-to-reel tape). Also, most modern home theatre processors that do have what are called ‘effects loops’ will **not** send audio from its phono input through the loop. In these cases, there is not a way to connect to SugarCube directly to these devices while using its built-in phono stage.

The solution is to add a phono stage (pre-amp) between your turntable and the SugarCube. Then connect the Line Out of the SugarCube to any available input on the integrated amp (except for the phono input, of course.)

2.7 **Connecting** Your SugarCube SC1-Plus

Connecting turntables with built-in phono pre-amps





For turntables with a built-in phono pre-amp, be sure to set the **PHONO/LINE** switch to **LINE** before plugging into the SugarCube.

Directly connecting the line-level output from your turntable.

1. Connect the turntable with a **built-in** phono pre-amp to **LINE IN.**
2. Be sure the turntable is set to **LINE** not **PHONO**.
3. Connect the **LINE OUT** of the SugarCube to any line-level input of your line stage, pre-amp, or integrated amp.

NOTE: SORRY TO BE A BROKEN RECORD, SO TO SPEAK, BUT DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.8 Starting Up Your SugarCube



(1) SugarCube just powered on



(2) SugarCube is ready to go (please expect about **three minutes to ready**.)



(3) Hold the 'Bypass' button for a few seconds to see LAN address (and WiFi SSID if using wireless.)

Please note: When you power up your SugarCube, it will take about **three minutes** to complete its startup procedures. During this time, the display will light up but remain blank. We apologize for the wait but we have some seriously sophisticated software loading and it simply takes additional time. When you hear the results of our efforts, you will agree that it was worth the wait.

Section 3

Networking the SugarCube



Part 1: Ethernet and WiFi
Part 2: Chrome Browsers
Part 3: Apple iOS
Part 4: Android



3.1 Using Ethernet or WiFi with SugarCube

3.1.1 **Pairing** your SugarCube



While setting up the connection between the SugarCube and the mobile or web browser SugarCube app, you will be prompted you enter a **Pair Code**, please use the code **1111**. (**Note:** you can push and release the **'Strength'** knob to obtain a unique pairing code but there is no need for this extra step, it remains to support legacy systems.)

3.1.2 **Connecting** Your SugarCube to a LAN or WiFi



If you would like to use the iOS or Android app or web browser to control your SugarCube SC1-Plus, please connect either the WiFi adapter supplied with your SugarCube or plug into a wired connection.

NOTES:

- THE SUGARCUBE SC1-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.

3.1.3 **Connecting** Your SugarCube via Wired Ethernet



- When connecting via wired Ethernet you need only open the iOS or Android App or follow a few steps to establish a browser connection.

NOTES:

- THE SUGARCUBE SC1-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.

3.1.4 **Connecting** Your SugarCube via WiFi



- If using the WiFi adapter you will either use the iOS or Android App to setup WiFi, or use Ethernet and a browser to configure the WiFi parameters.

NOTES:

- THE SUGARCUBE SC1-PLUS BASIC OPERATIONS ARE FULLY FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT THE NEED TO CONNECT TO A NETWORK.
- SOFTWARE UPDATES ARE AUTOMATIC AND PROVIDED WHEN THE SUGARCUBE IS CONNECTED TO A WIFI OR ETHERNET NETWORK.



3.2 Using a Browser with SugarCube

3.2.1 **Controlling** the SugarCube from a Web Browser

Finding the IP address of your SugarCube SC1-Plus



1

Plug in an Ethernet Cable



2

Press and **hold** 'BYPASS' for 3 to 5 seconds

Your SugarCube will then enter a display mode where it will show you its IP address, among other information.

3

You will use an address that looks like this example. It usually will start with 192 or with 10.

3.2.2 **Controlling** the SugarCube from a Web Browser

Entering the IP address with addition information into the browser

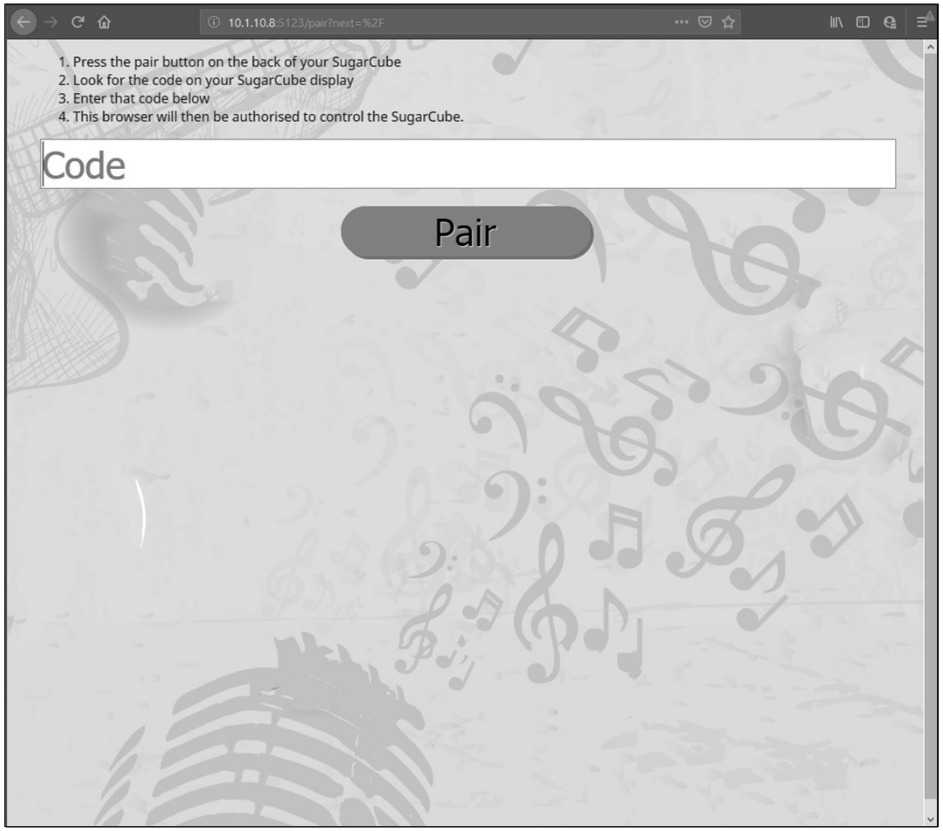


Important Note: **the SugarCube web interface is compatible with Google Chrome, Microsoft Edge, & Apple Safari web browsers. We cannot guarantee compatibility with any other brand.**

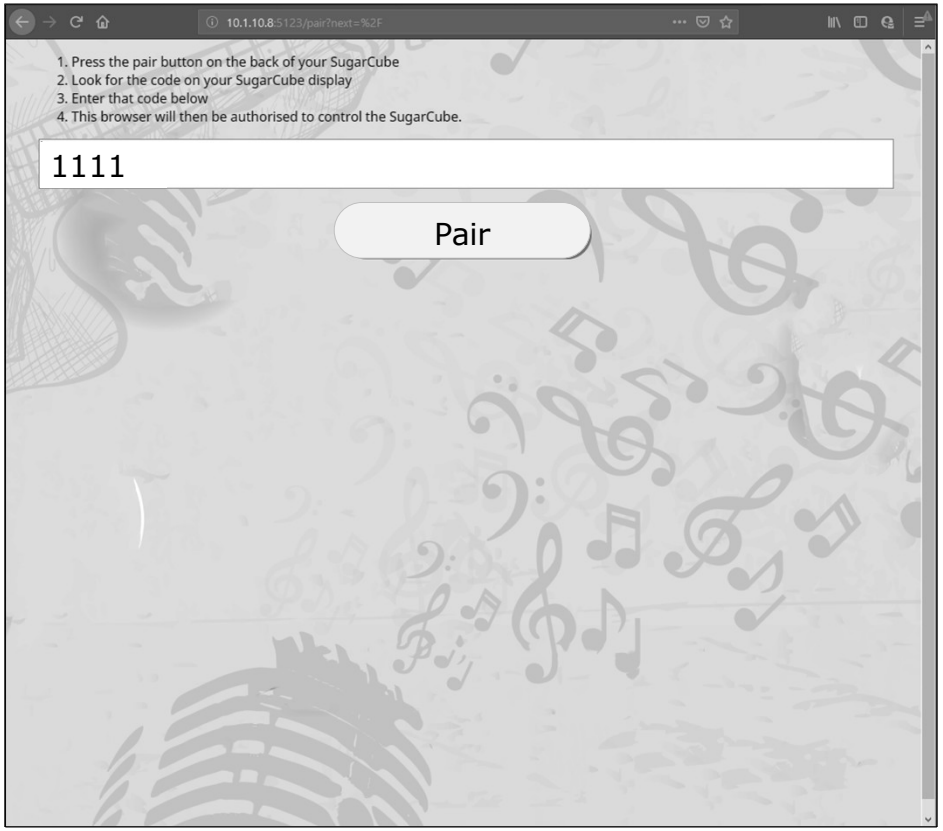
(1) Type the SugarCube's IP address into the **address bar** then add a colon ':' and '5123'. PLEASE NOTE: the '5123' is a fixed port number and NOT the serial number of your SugarCube.

(2) Press 'Enter' and the SugarCube webpage will appear in the browser.

3.2.3 **Controlling** the SugarCube from a Web Browser



(1) You will see a screen appearing in your browser to accept a pairing code.



(2) The code '1111' will appear automatically, if there is only one SugarCube on your network.

The Main Menu will automatically appear in one to two seconds after this.

3.2.4 **Controlling** the SugarCube from a Web Browser

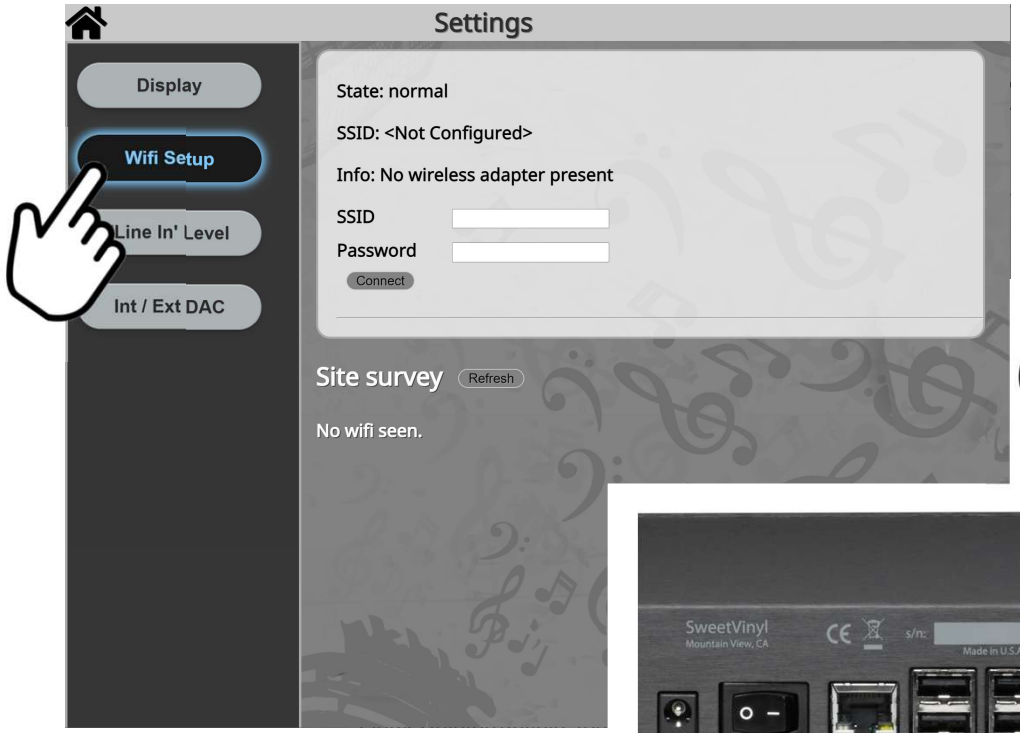


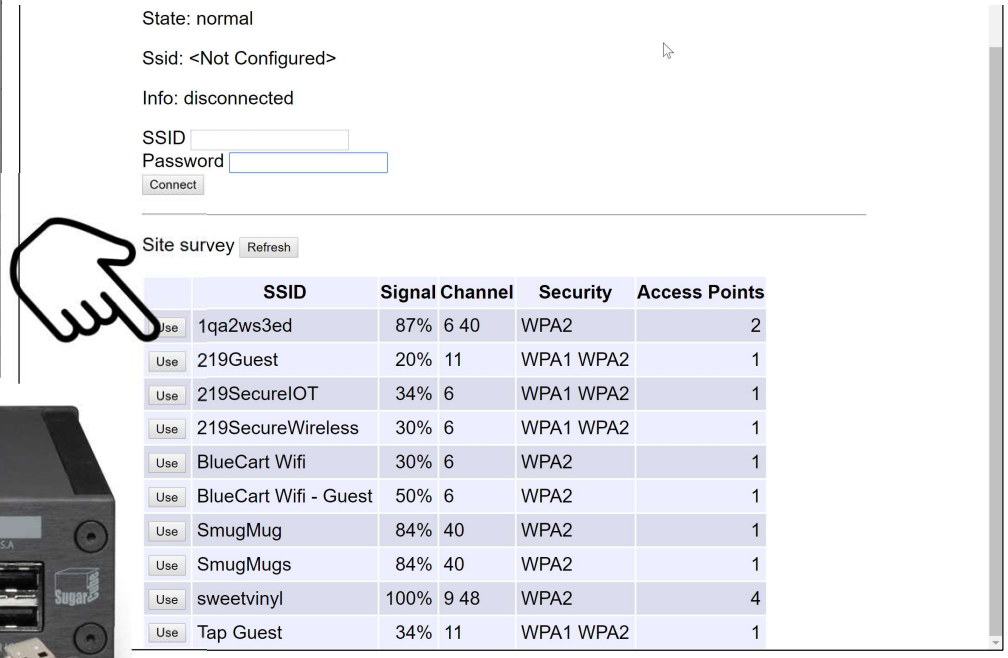
Important Note: **the SugarCube web interface is compatible with Google Chrome, Microsoft Edge, & Apple Safari web browsers. We cannot guarantee compatibility with Firefox.**

The SugarCube user interface as it appears in a browser.

3.2.5 **Controlling** the SugarCube from a Web Browser

Setting up Wi-Fi (in place of wired Ethernet), 1 of 4







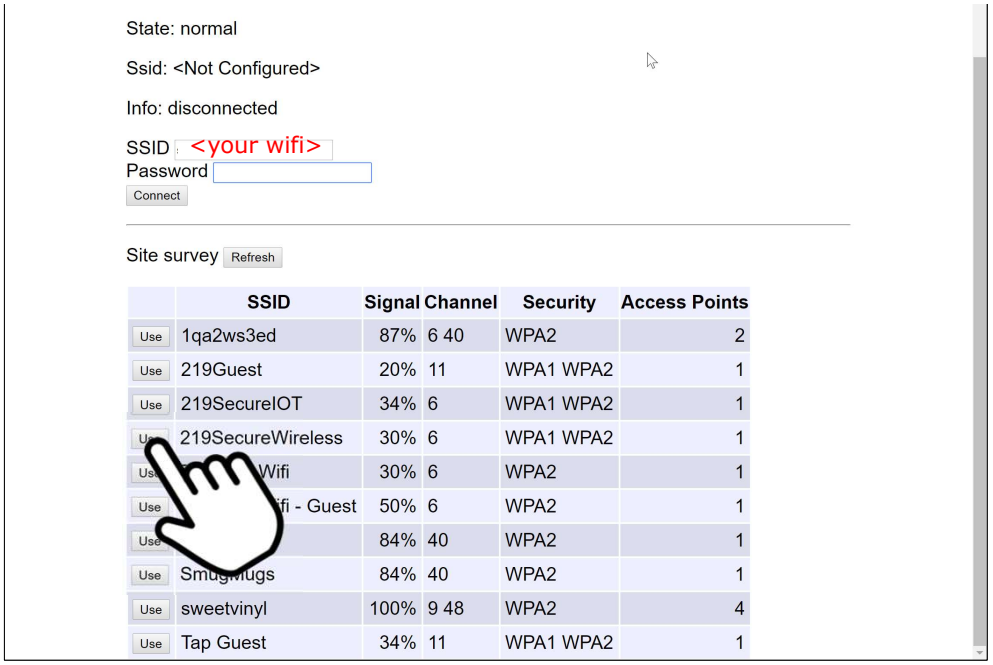
(1) If wired Ethernet is not available in your sound room, then take the SugarCube to your router just for these WiFi setup steps.

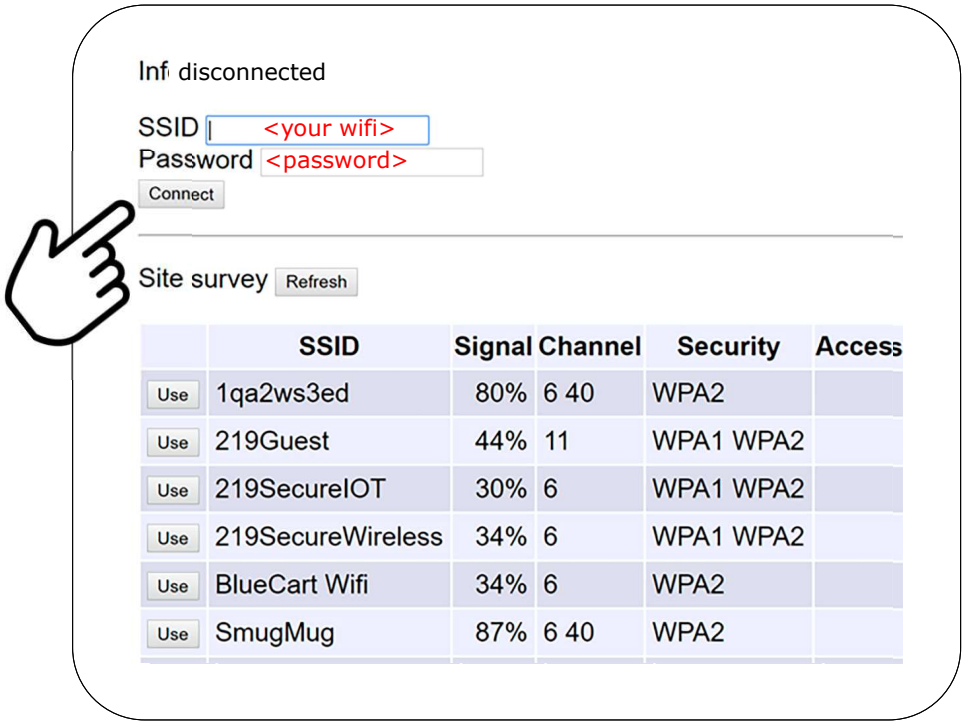
(2) You need both Ethernet and the WiFi stick plugged into the SugarCube.

(3) From the '**WiFi Setup**' menu, find your Wi-Fi name on the list of SSIDs.

3.2.6 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 2 of 4





(4) Please select your network by clicking on the 'Use' button next to your SSID.

(5) Next, enter the password for your WiFi then click on 'Connect.'

3.2.7 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 3 of 4

Press the pair button on your SugarCube for at least 5 seconds to start the setup wifi network.

State: connect

Ssid: s\ <your wifi>

Info: connecting (getting IP configuration)

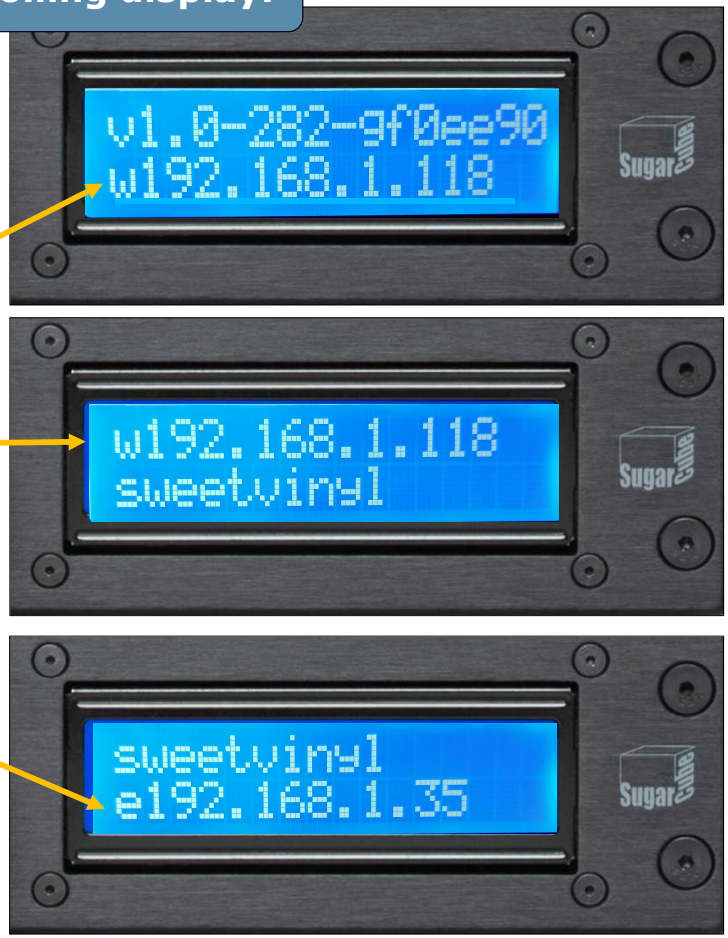
SSID sw <your wifi>
Password <password>
Connect

Site survey Refresh

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIOT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1

(6) The system will indicate that it is connecting to your Wi-Fi.

2 line scrolling display:

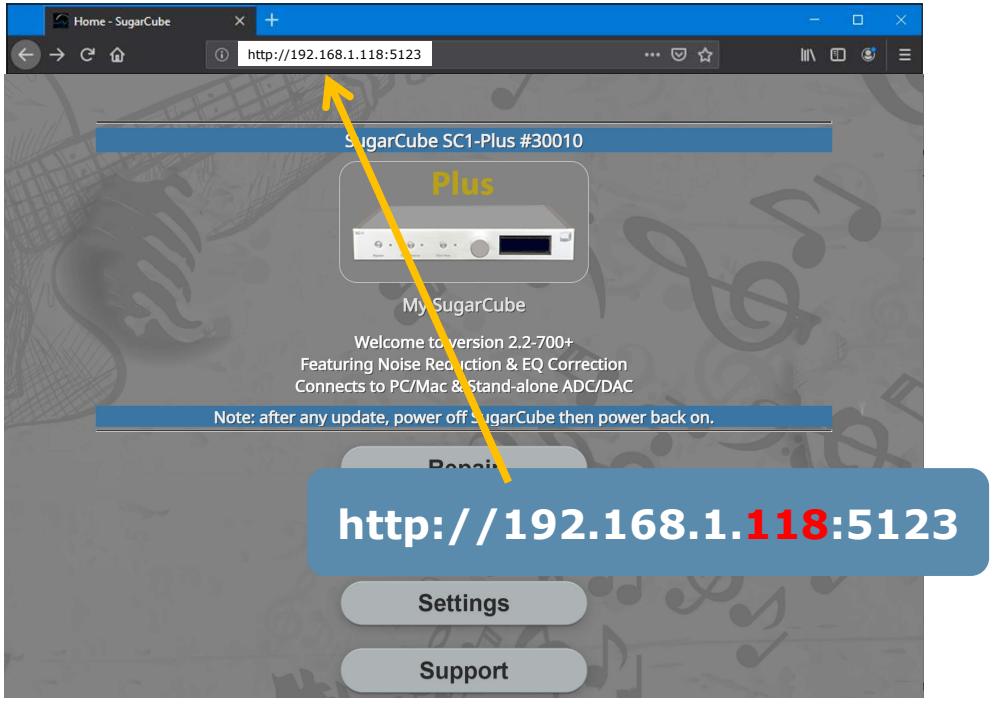


(7) Once connected, you will now see, scrolling by, both your new Wi-Fi address (w) and your currently wired Ethernet IP address (e). Remember, you are currently communicating with the SugarCube via the wired IP address.

3.2.8 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 4 of 4





(9) So, when you disconnect the wire, your browser will no longer communicate. You will have to change the IP address in your browser, in this example, to the Wi-Fi address ending in '.118'. (Once the wire is disconnected, the front panel display will show only the WiFi IP address.)

(10) Usually, you need only to change the last group of numbers and press 'Enter', and you should reconnect (sometimes you may have to re-enter the entire address).

(11) Please note, if your SugarCube cannot connect to your WiFi, contact support@sweetvinyl.com with the details of your system (WiFi make and model and the serial number of your SugarCube.)

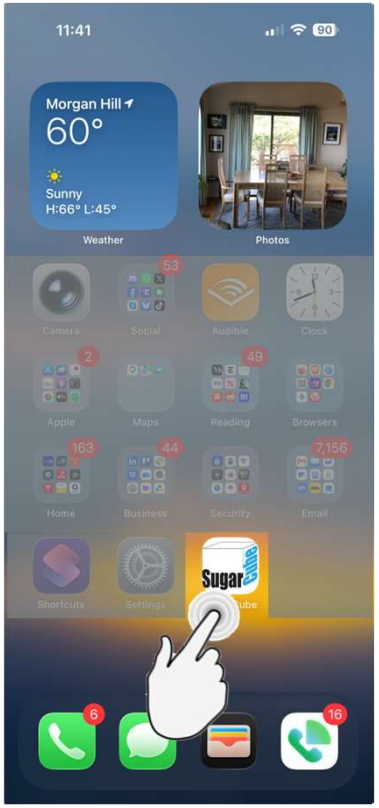


3.3 Using Apple iOS App with SugarCube

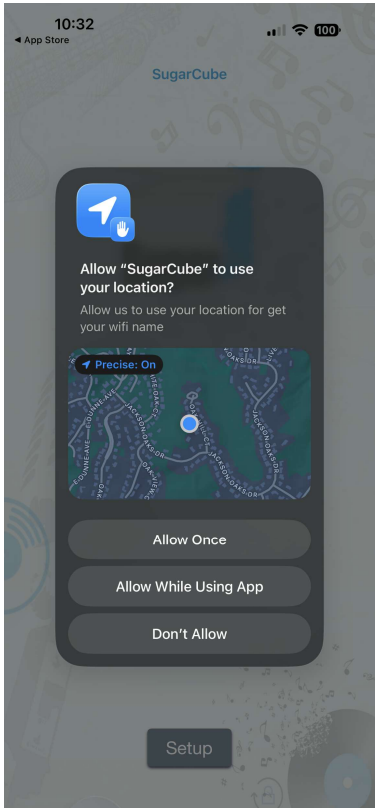
3.3.1 **Wi-Fi** Setup on iOS device

1. Using iPhone or iPad.
2. From the Apple App Store, download the '**SugarCube**' app to your Apple mobile device. (Look for: **SugarCube by SweetVinyl**.)
3. Connect the WiFi dongle supplied with your SugarCube to one of the USB-A ports on the back panel.

3.3.2 **Wi-Fi** Setup on iOS device



(1) Launch SugarCube app



(2) If first time using the app, allow locations services by selecting '**Allow While Using App**'.



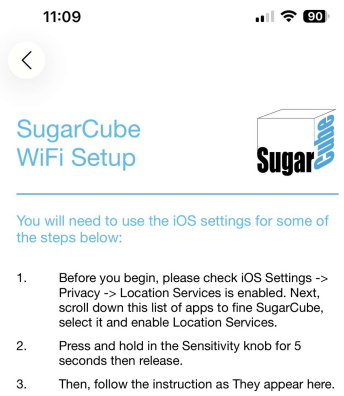
(3) App is searching for SugarCubes that may already be setup on the LAN if any.

For the first time user, nothing will be found, of course.



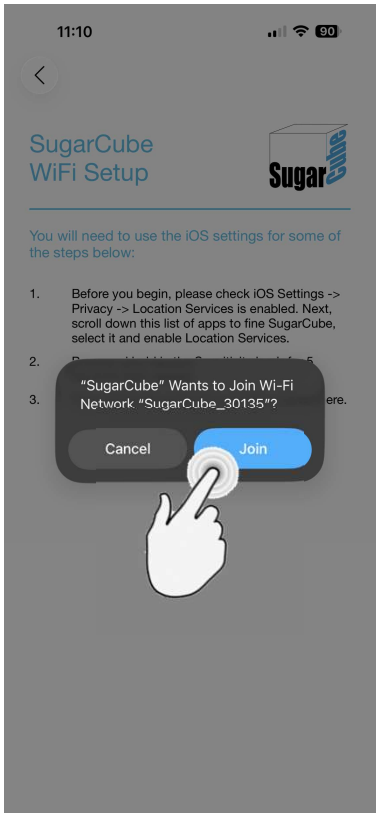
(4) When no SugarCube is found, press **WiFi Setup** to initial the setup process.

3.3.3 **Wi-Fi** Setup on iOS device



(5) The user is instructed to press and hold the front panel knob for **5 seconds** to put the SugarCube into '**WiFi Setup Mode**'.

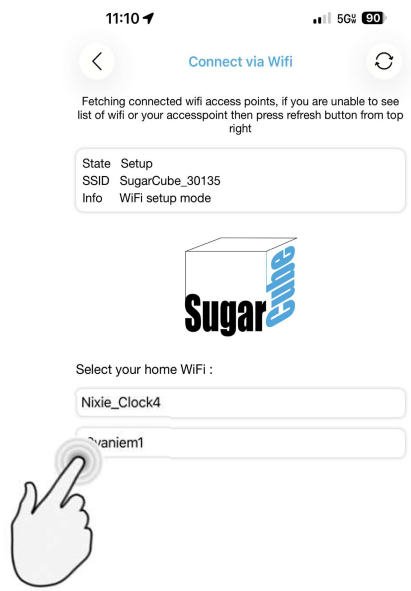
Confirmation is displayed on the front panel.



(6) In WiFi Setup mode, after a few seconds, the SugarCube switches the Wi-Fi adapter into becoming a **Wi-Fi access point** that the iPhone will then be asked to join.



(7) After joining the SugarCube's Wi-Fi service, the app searches the home for any available Wi-Fi access points. This can take ten to twenty seconds.

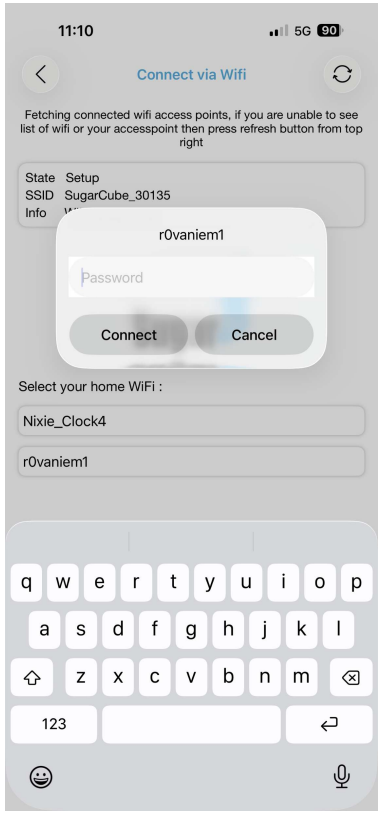


(8) When the sweep is complete, the app displays the available access points.

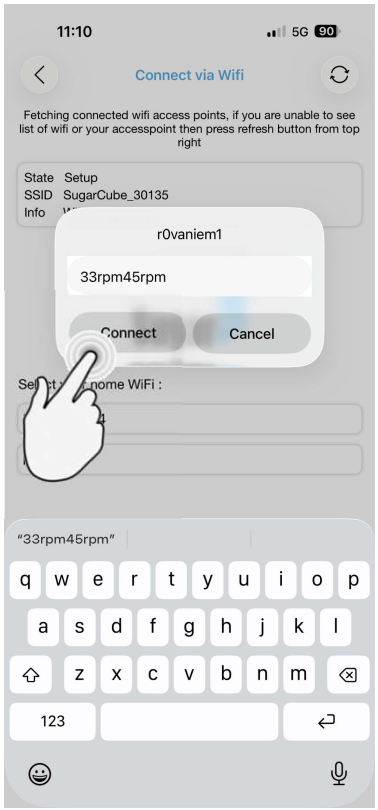
Click on your desired network.

Apple iOS

3.3.4 **Wi-Fi** Setup on iOS device



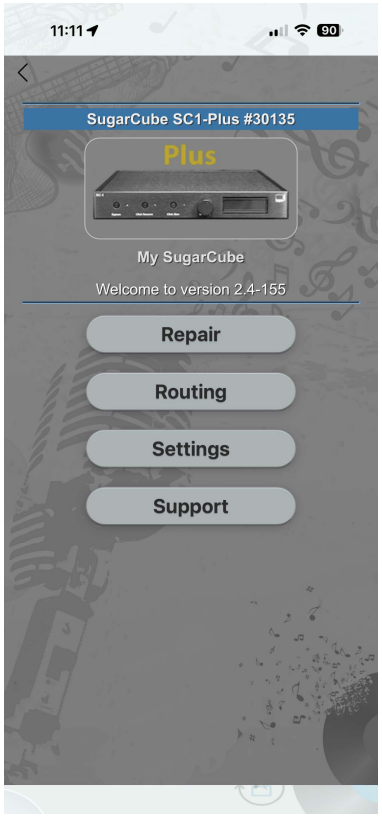
(9) Upon selecting a Wi-Fi network to join, a window will appear to enter the password.



(10) After entering the password, press '**Connect**' and you are done.



(11) The app is now completing the connection and preparing to load the SugarCube menu.



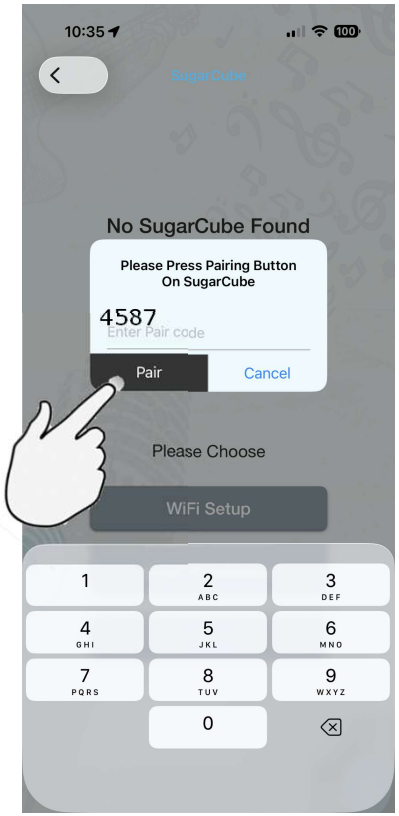
(12) The SugarCube main menu will appear automatically every time you load the app on your phone.

Apple iOS

3.3.5 **Wi-Fi** Setup on iOS device

If you have more than one SugarCube on your network, you will be asked for a pairing code at different times during setup.

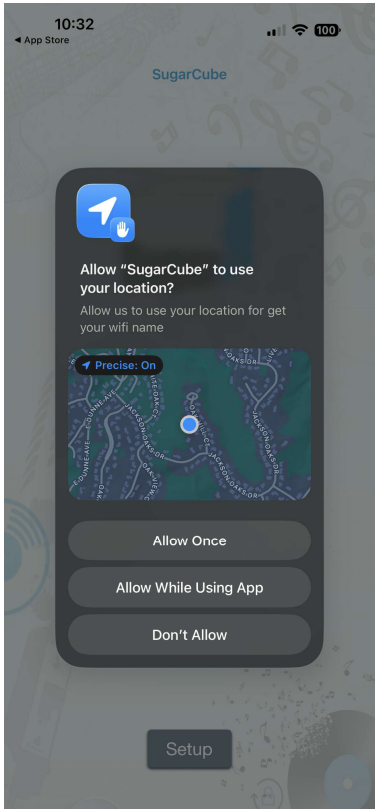
At any time, when asked for a pairing code, press and release the front panel button and enter the code from the front panel display to connect with that SugarCube.



3.3.6 Connecting first time on Wired Ethernet



(1) App is loading



(2) First time use requires permission to use locations. We recommend selecting 'Allow While Using App'.



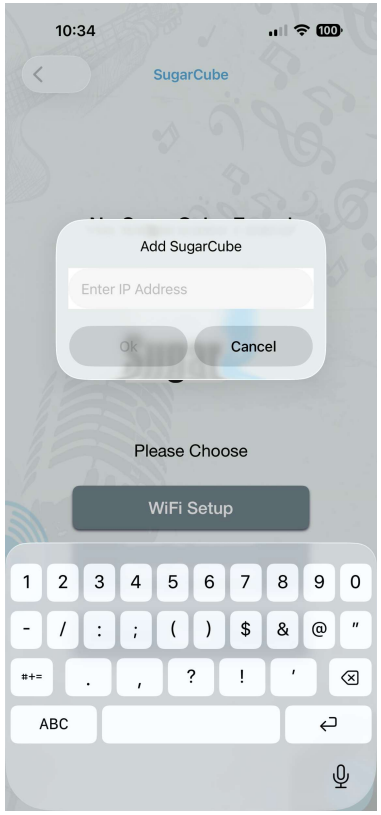
(3) Select 'Enter IP Address' to access SugarCube on wired Ethernet.



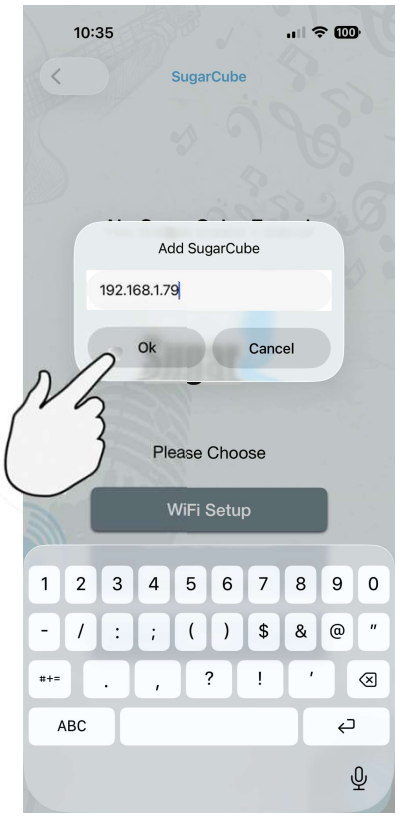
(4) App loading setup page.

Apple iOS

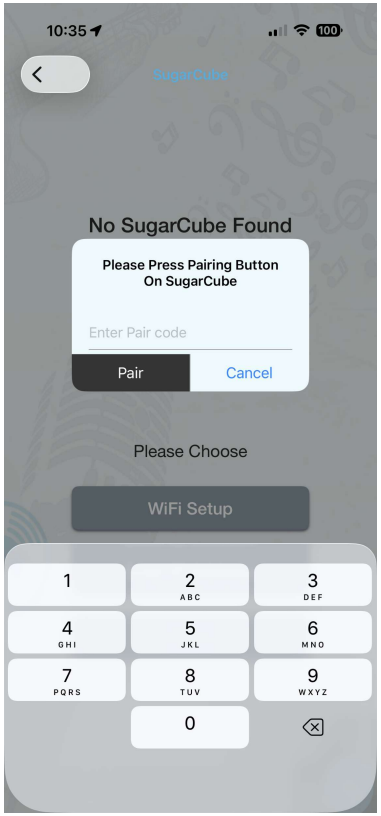
3.3.7 **Connecting** first time on **Wired Ethernet**



(5) Enter IP Address in window.



(6) Press 'Ok'.



(7) If only one SugarCube on your network, just enter '**1111**'

If more than one, enter code displayed on front panel found by pressing '**Select**' knob.



(8) Click '**Pair**' to get to the SugarCube Main Menu.

Apple iOS

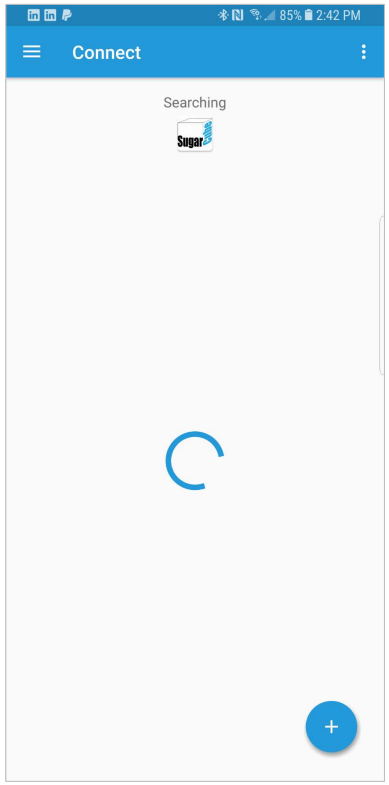


3.4 Using Android App with SugarCube

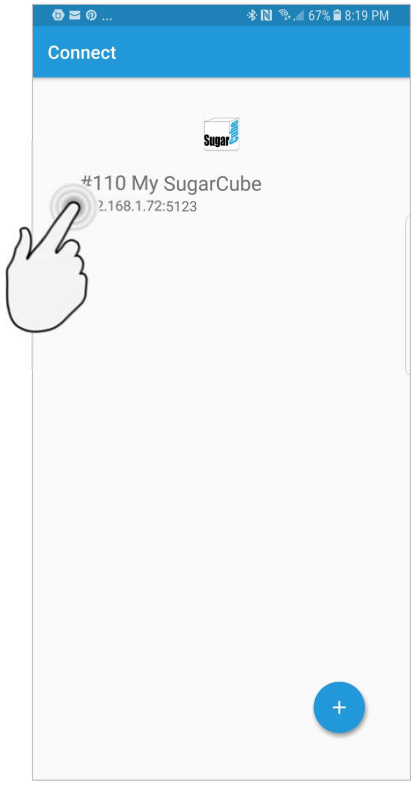
3.4.1 **Connecting** with SugarCube on **Wired Ethernet**



(1) App is loading



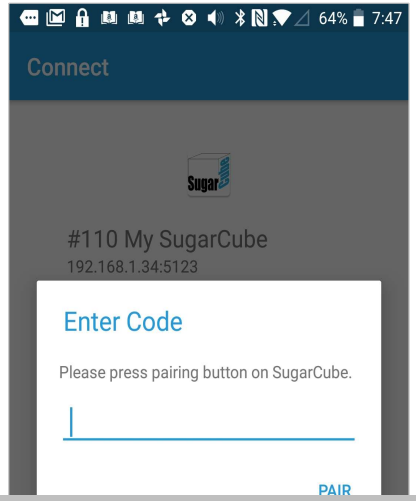
(2) App is searching for SugarCubes on the LAN



(3) App has found a SugarCube and will now initiate a connection.

Android

3.4.2 **Connecting** with SugarCube on **Wired Ethernet**

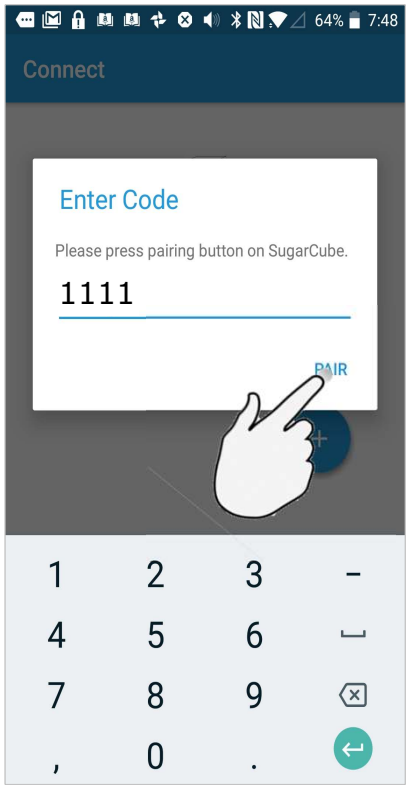


For pairing, enter:

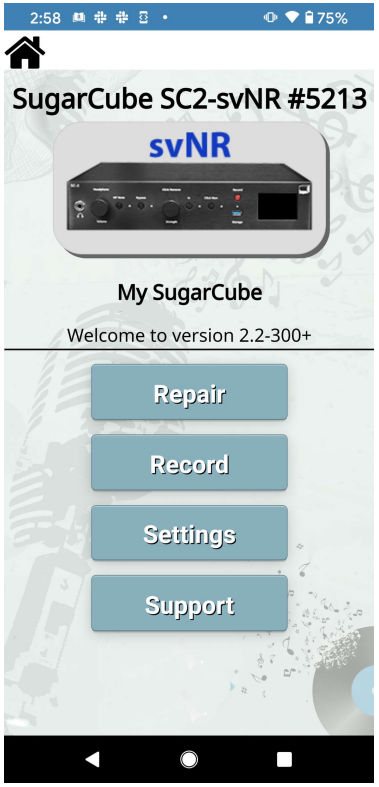
1 1 1 1

as a pairing code
(you may still get a code from your SugarCube, if you with).

(4) The app needs you to enter a pairing code, press the knob on the front panel.



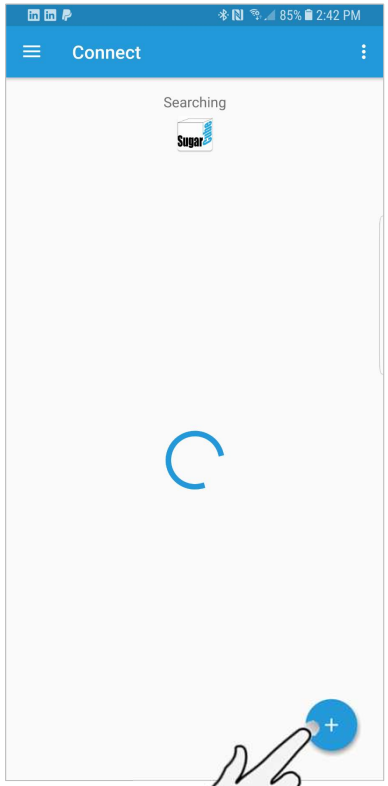
(5) App needs you to enter the pair code '**1111**' or what is displayed on the front panel display then touch '**PAIR**'.



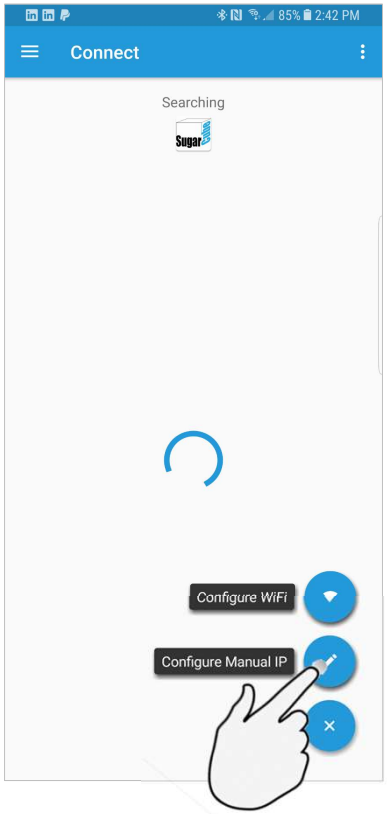
(6) App is linked to SugarCube and ready for use.

Android

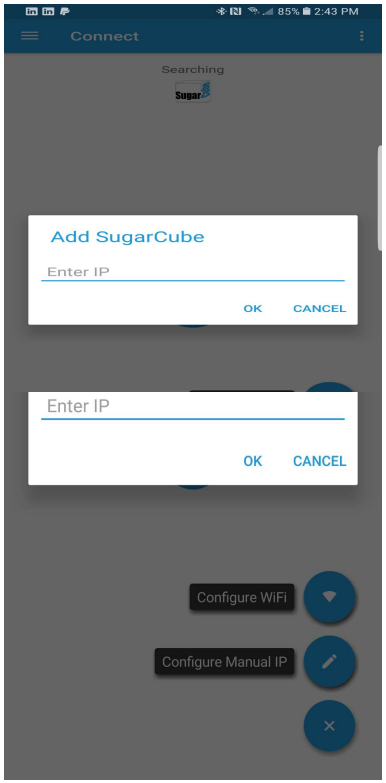
3.4.3 **Connecting** with SugarCube Using **IP Address**



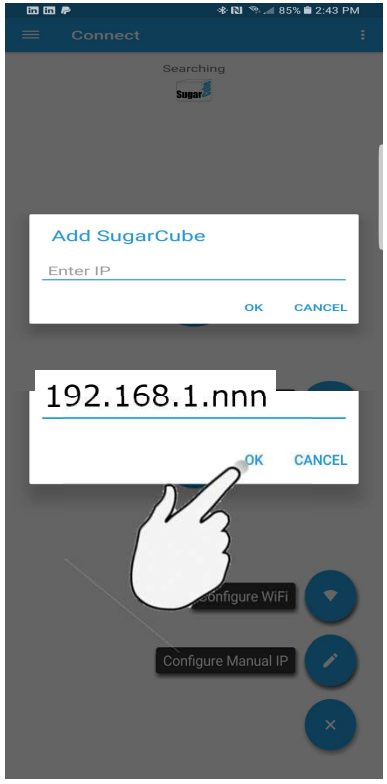
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.



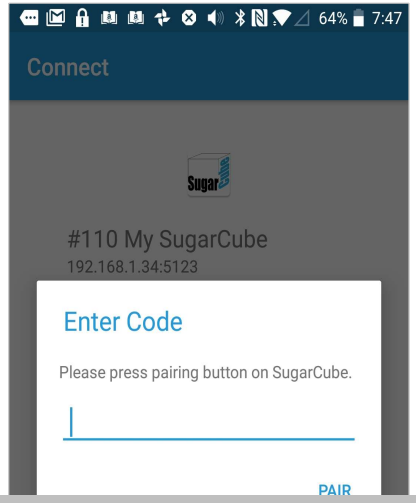
(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.



(4) Enter the IP address as displayed on the front panel display.

Android

3.4.4 **Connecting** with SugarCube Using **IP Address**

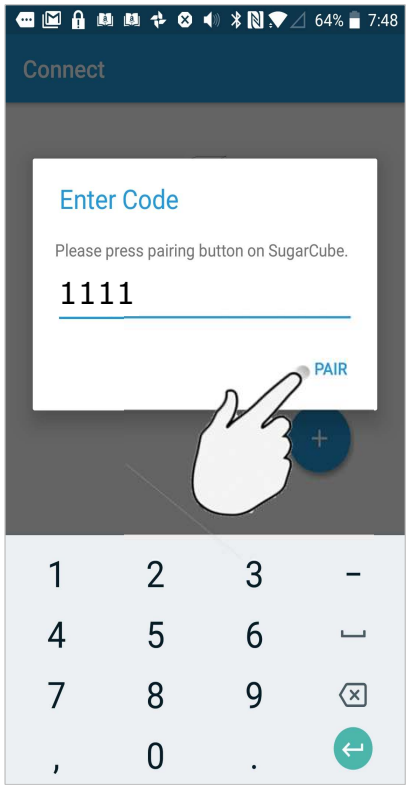


For pairing, enter:

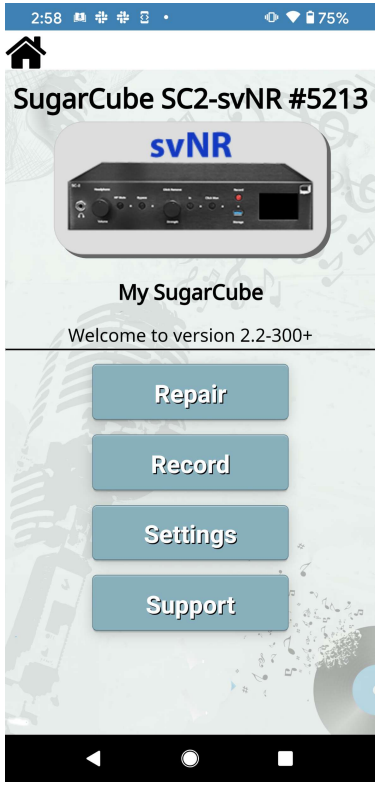
1 1 1 1

as a pairing code
(you may still get a code from your SugarCube, if you wish).

(5) The app need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.



(7) App is linked to SugarCube and ready for use.

Android

3.4.5 **Wi-Fi** Setup on Android device

You will need:

1. An Android phone or tablet.

2. To connect with WiFi on your Android device on the same network that you plan to use with your SugarCube with WiFi or wired Ethernet.

3. Download the Android App from the Google Play Store.
   (Look for: SugarCube by SweetVinyl.)

4. To plug in the WiFi dongle supplied with your SugarCube or a wired Ethernet cable from a port on your router.

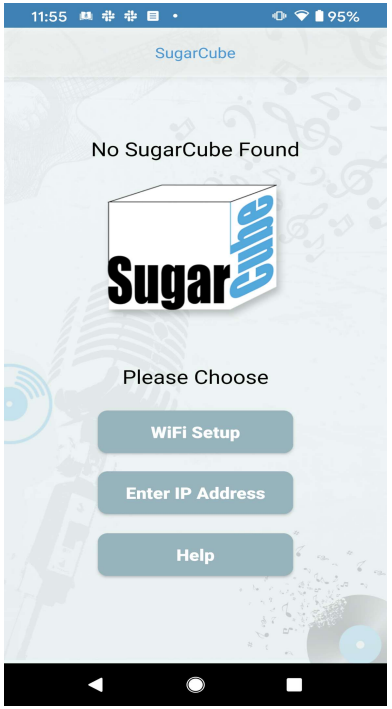
3.4.6 **Wi-Fi** Setup on Android device



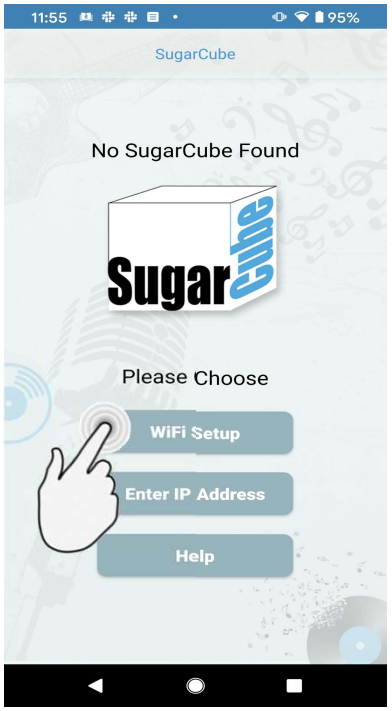
(1) App is loading



(2) App is searching for SugarCubes on the LAN

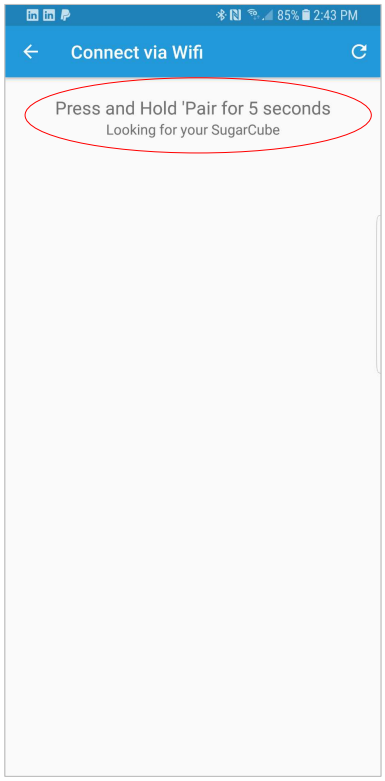


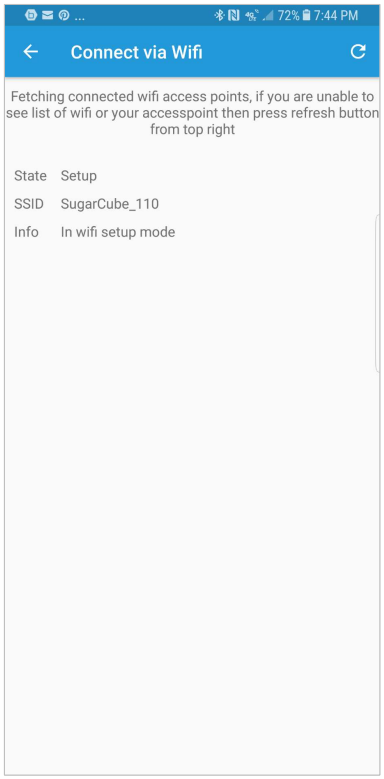
(3) No SugarCube is found so the app offers the option to run WiFi setup.

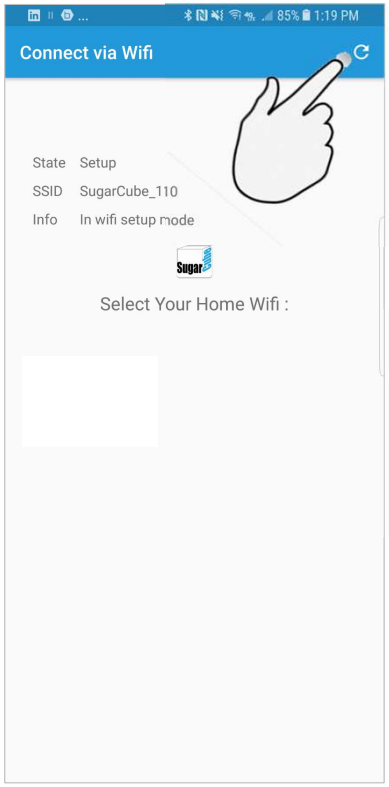


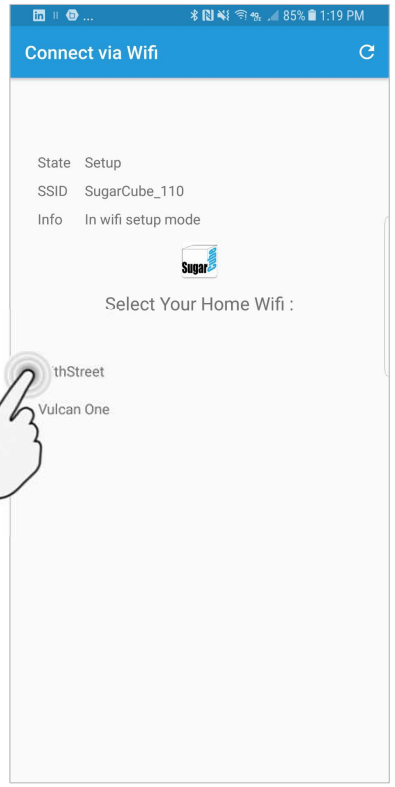
(4) User taps ‘WiFi Setup’ to start the process.

3.4.7 **Wi-Fi** Setup on Android device









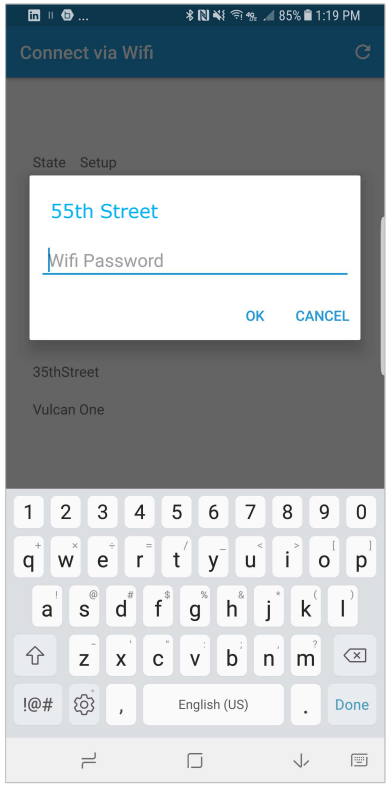
(5) Next, press and hold the pairing button (on the rear panel) for 5 seconds as instructed.

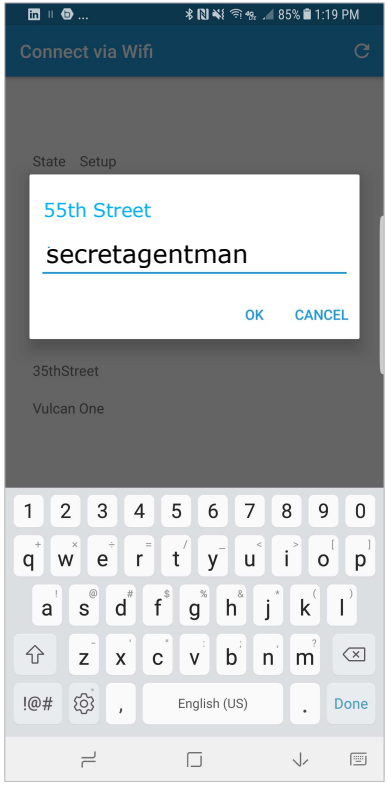
(6) After this, your SugarCube will be configured as an access point and display its SSID (e.g. SugarCube_110) In this screen the app has now connected to it automatically.

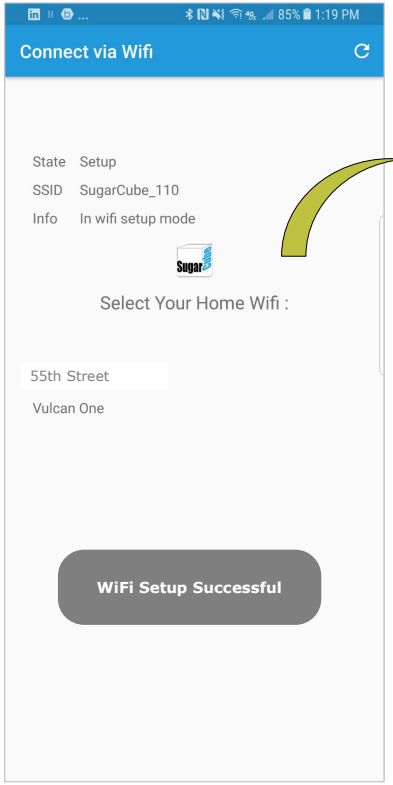
(7) Next, the SugarCube will find your available WiFi connections. This could take a minute but if nothing shows, please press the refresh (upper right).

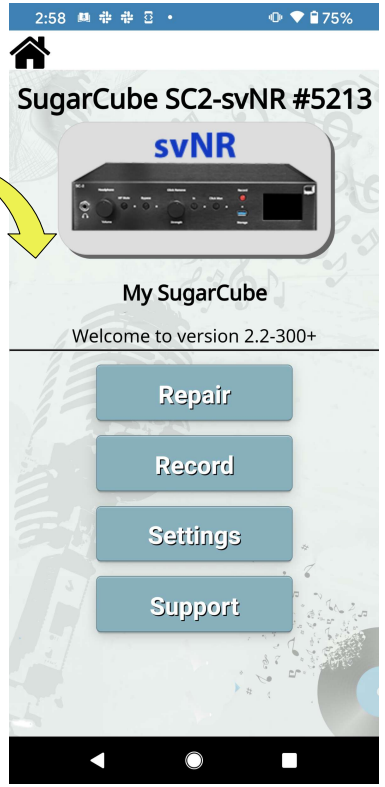
(8) When you see your network, select the desired connection.

3.4.8 **Wi-Fi** Setup on Android device









(9) Next, enter the password for your WiFi network.

(10) The app has connected successfully with the SugarCube and will now jump to the main screen.

(11) The connection process is complete and you are ready to go.

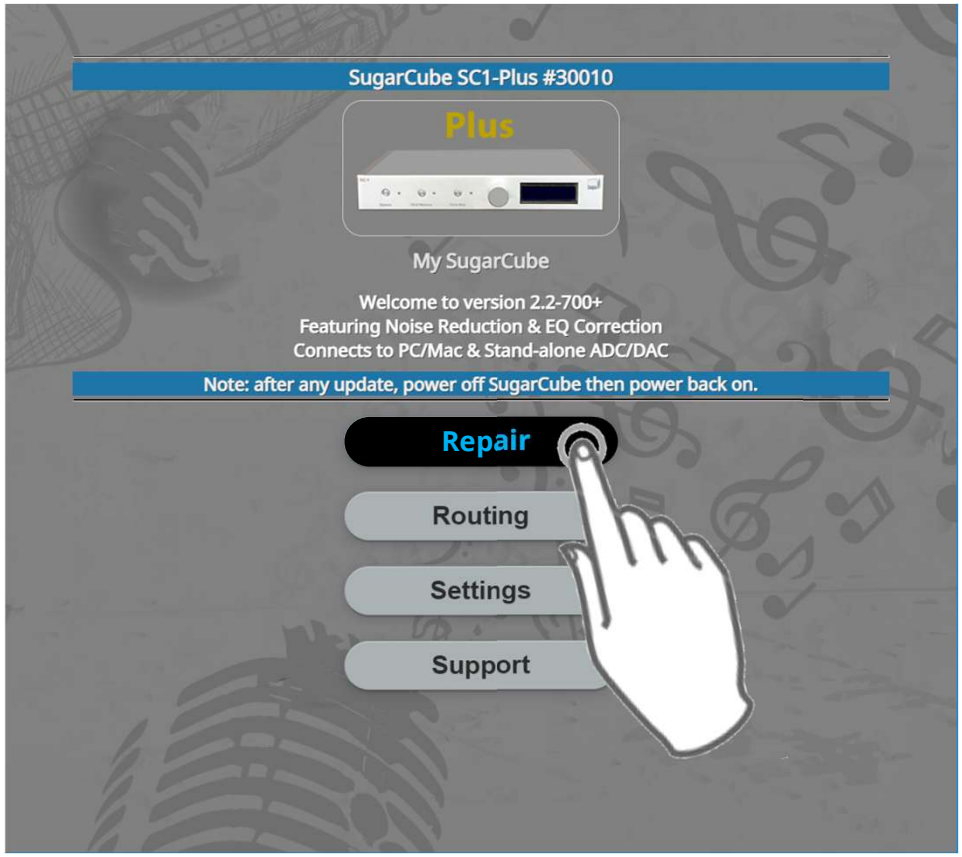
Section 4

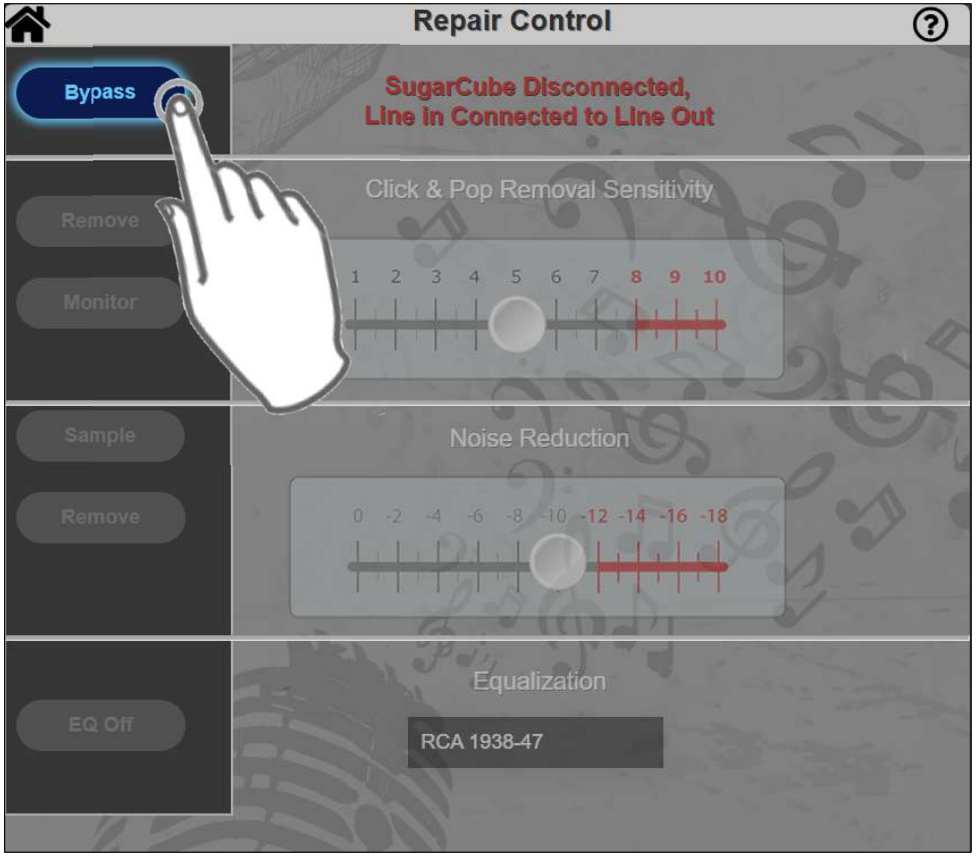
Controlling SugarCube via a browser or app

Chrome, Edge, or Safari browser &
iOS or Android app



4.1 Home Page and Repair Control



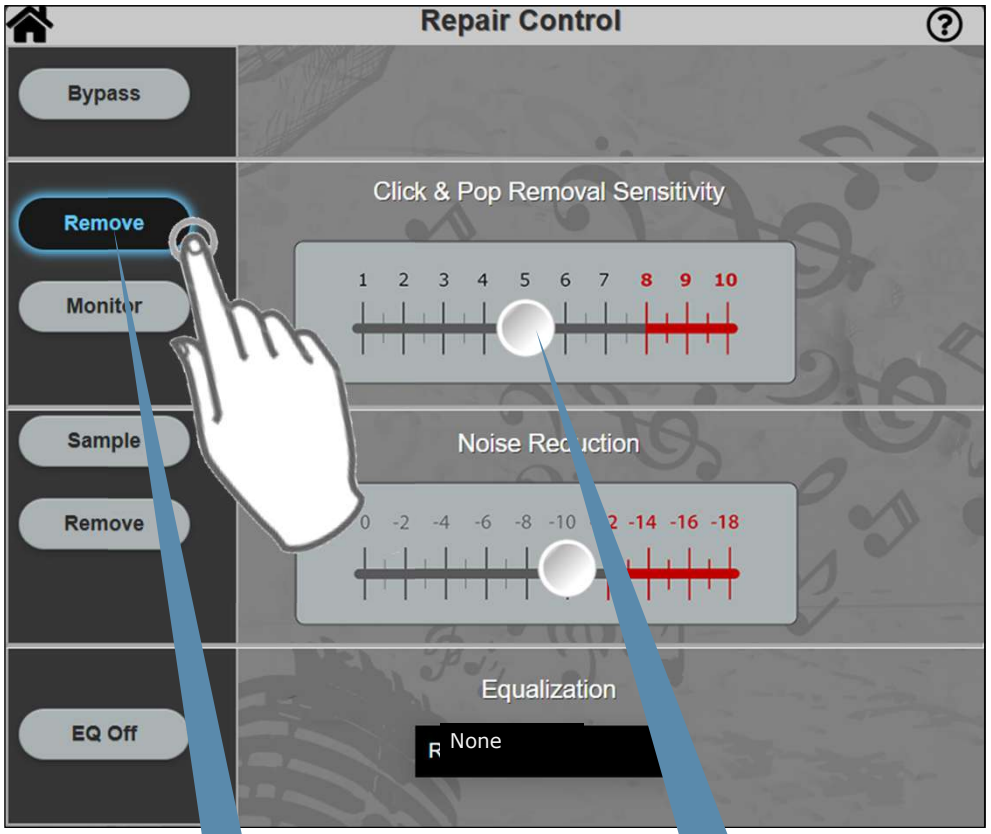


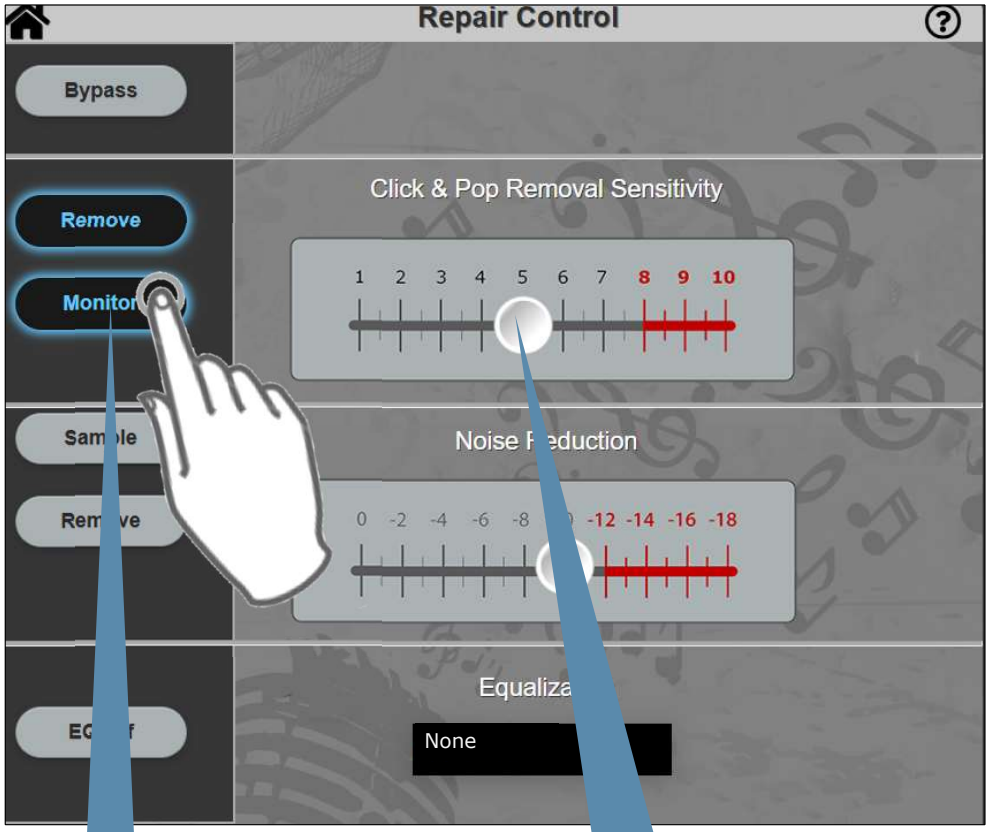
(1) Click on 'Repair'

(2) Click and Pop and Bypass functions available mirror the front panel controls. A change here is reflected on the front panel and a change on the front panel changes the above.

4.2 **Repair Control** Functions

Controlling Click and Pop Removal





The Click and Pop removal function has been engaged.

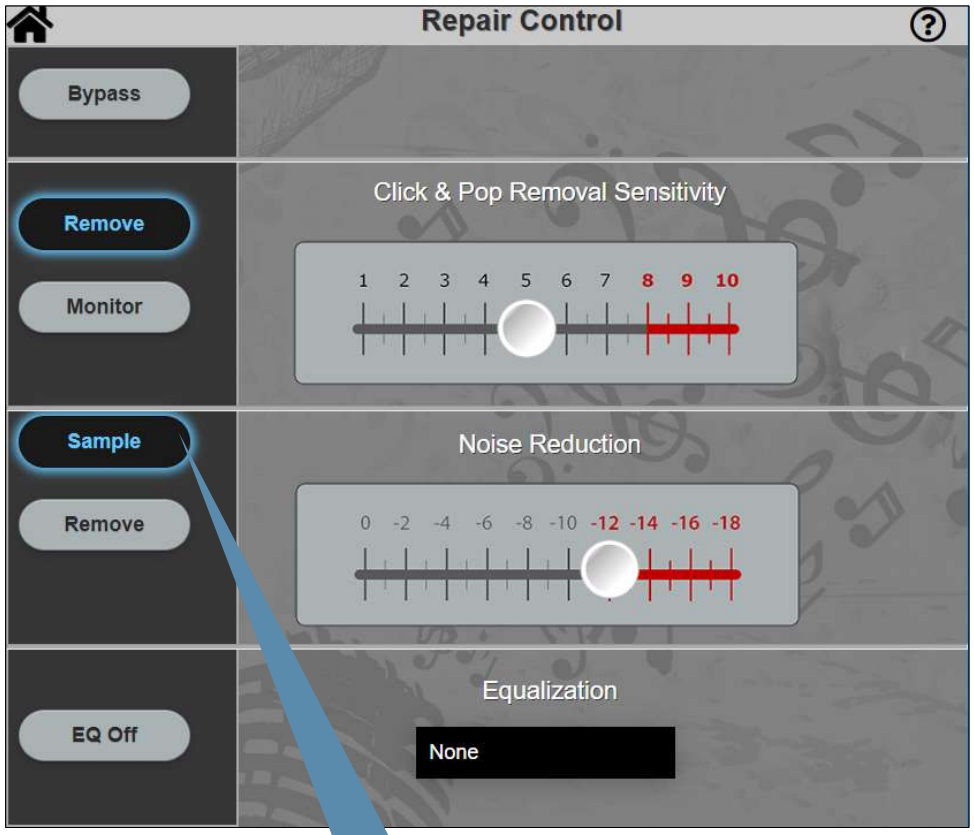
Adjust the slider to listen to the result of setting 'Strength'.

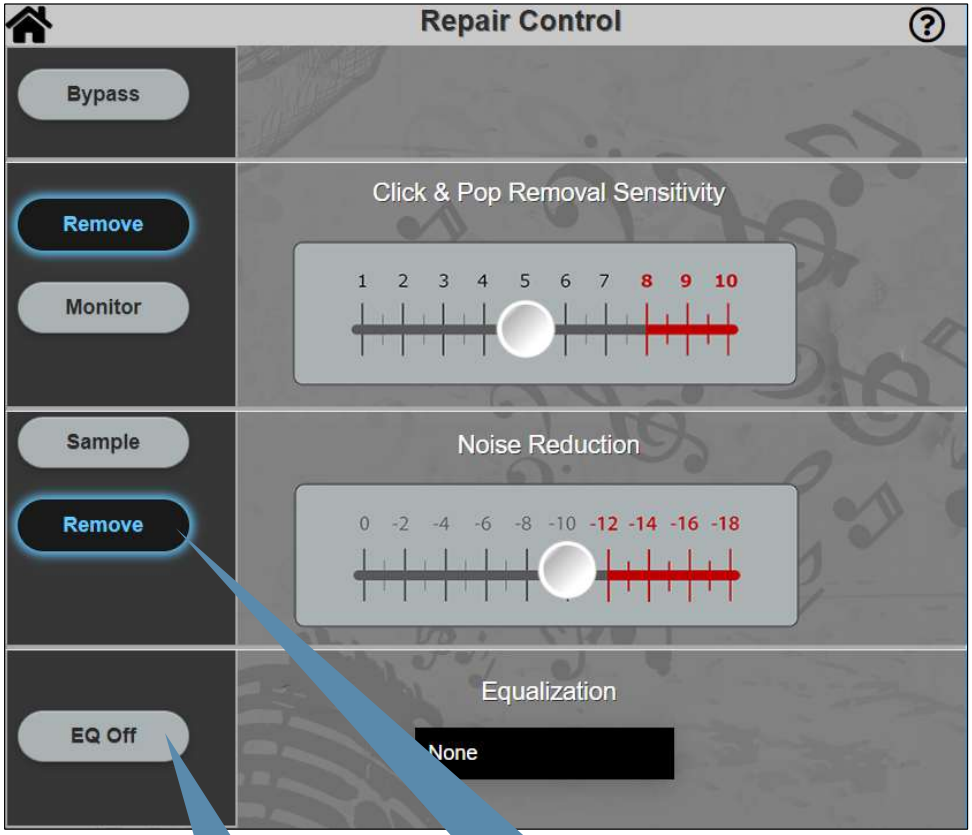
Monitor the clicks and pops that are being removed from the audio passing through the SugarCube.

Setting of **'4'** or **'5'** is all you need for all but the most damaged LPs

4.3 Noise Reduction Functions

Controlling Noise Reduction





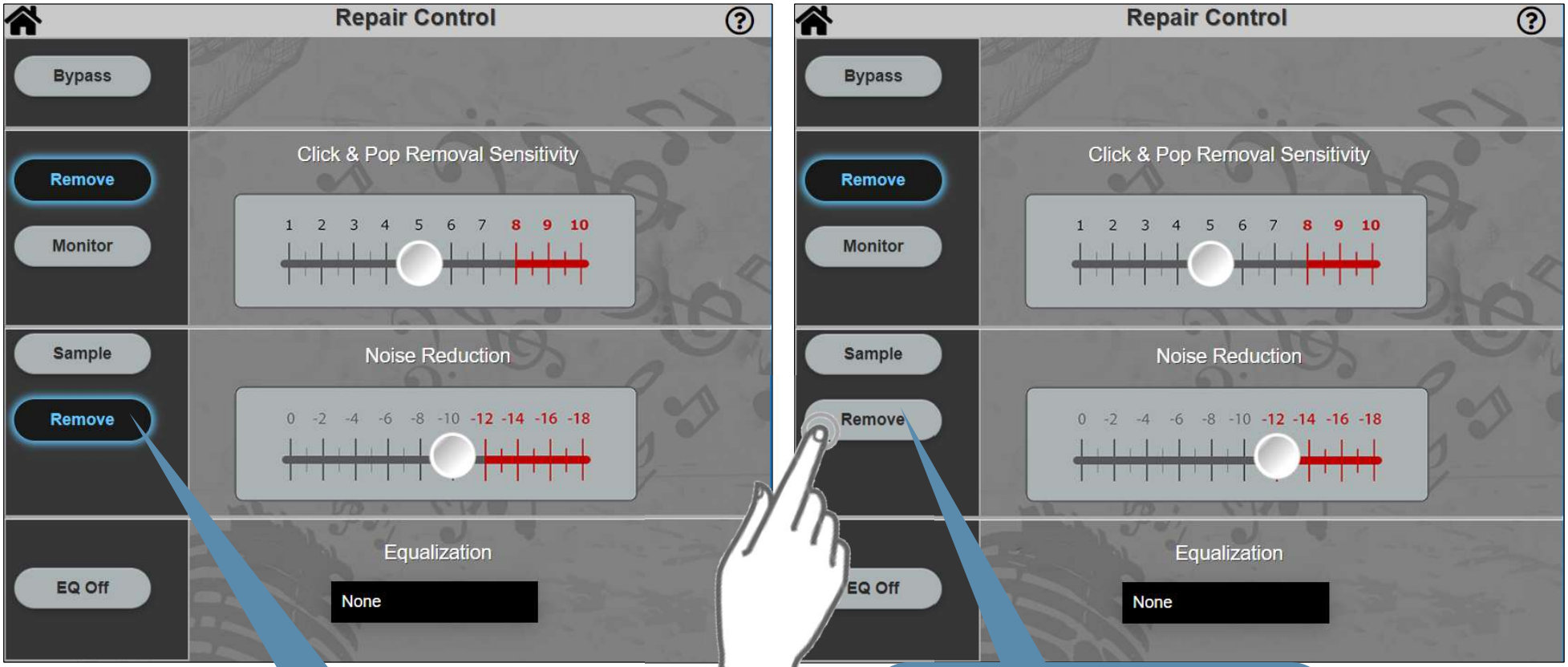
Click on '**Sample**' to initialize '**Noise Reduction**'. Drop the needle into the lead-in groove and the system takes a three second sample of vinyl noise from the lead in groove. When done, '**Sample**' turns off by itself.

Note: **Record Equalization (EQ)** is off (in this example) and can be turned on at any time.

'**Noise Reduction**' engages/disengages noise reduction (note: setting the slider to '0' also disengages Noise Reduction.)

4.4 Noise Reduction Functions

Controlling Noise Reduction

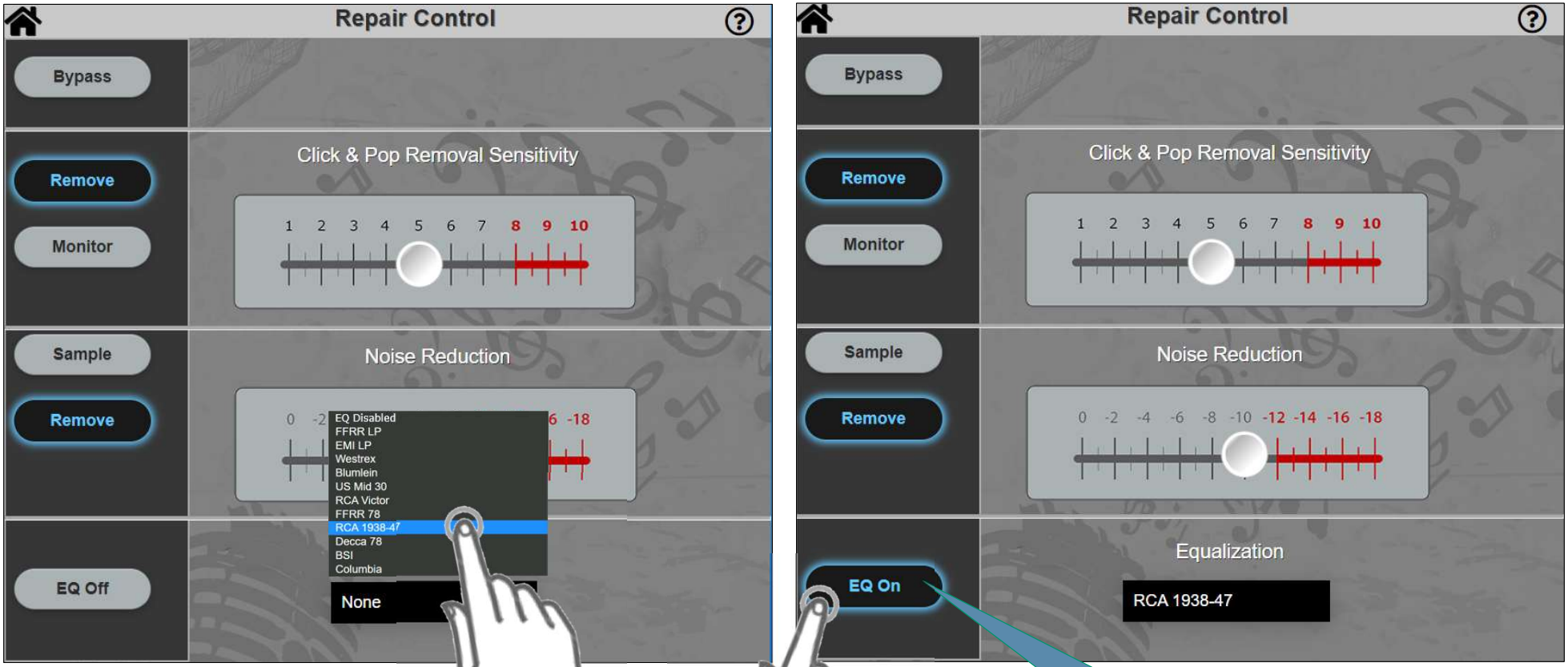


NR can be dis-engaged any time by clicking here or moving the slider to '0'.

Clicking on '**Remove**' disengages the NR process but you do not need to re-sample if the next vinyl LP is from the same time period and label.

4.5 Equalization (EQ) Compensation

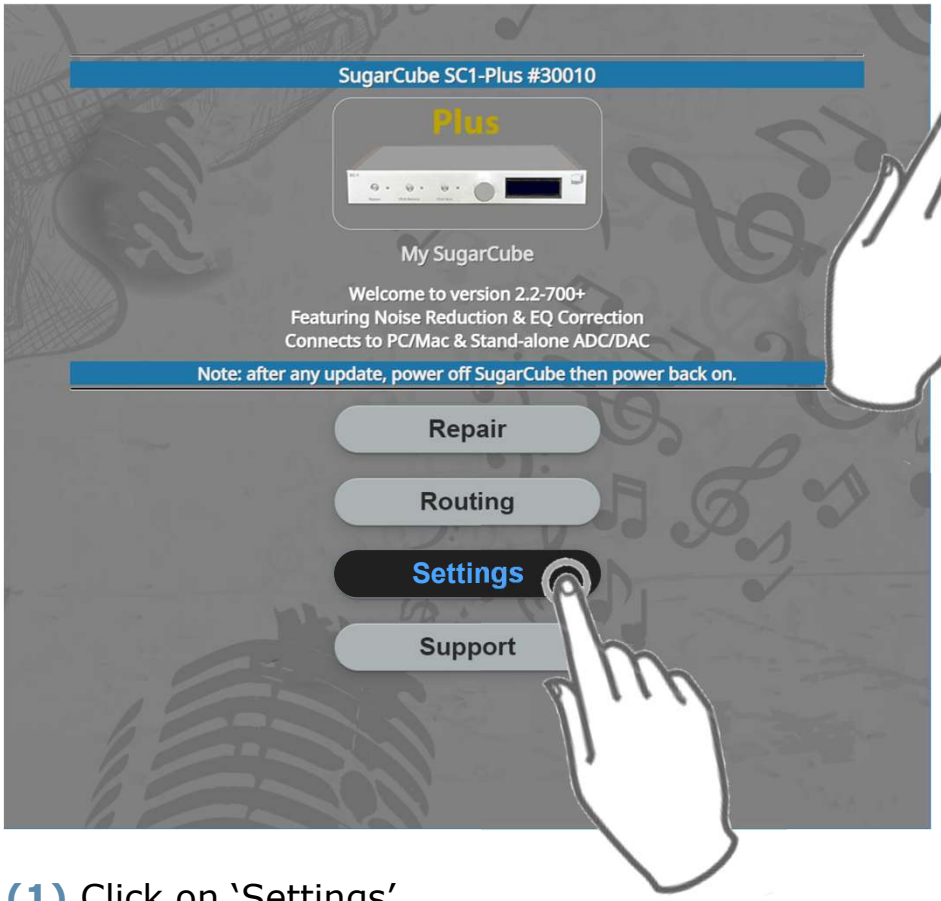
Controlling Playback EQ Curves



The desired EQ curve can be chosen from the pop-up menu.

Clicking on EQ will engage it at any time.

4.6 **Settings** and **Display** Functions

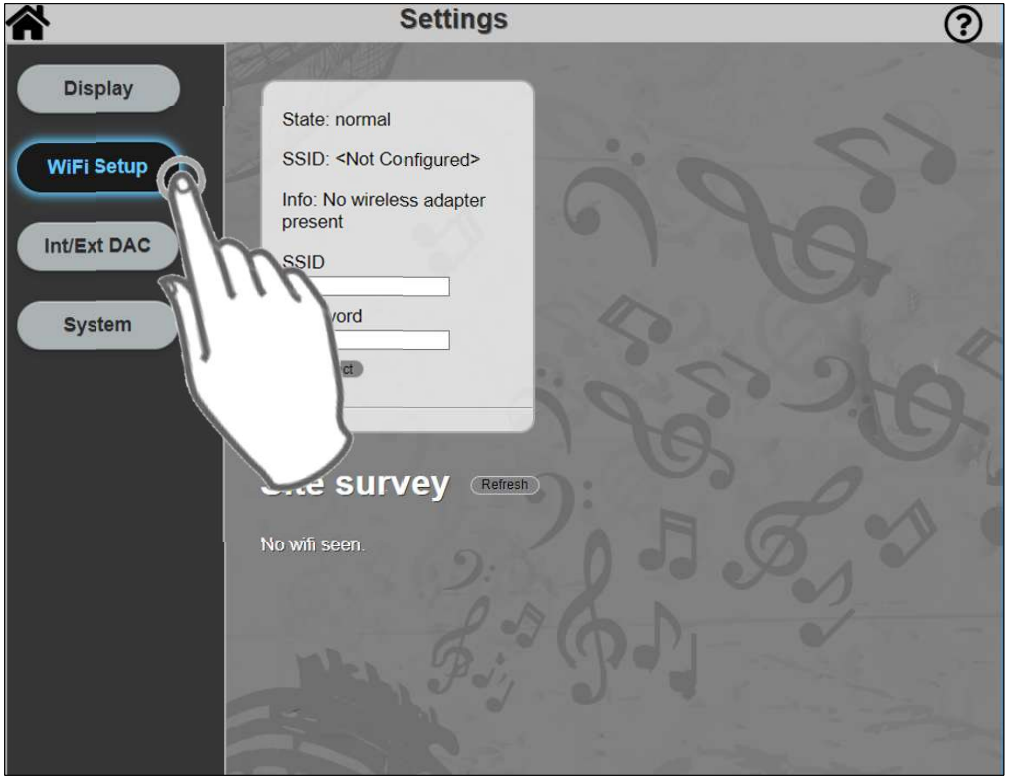


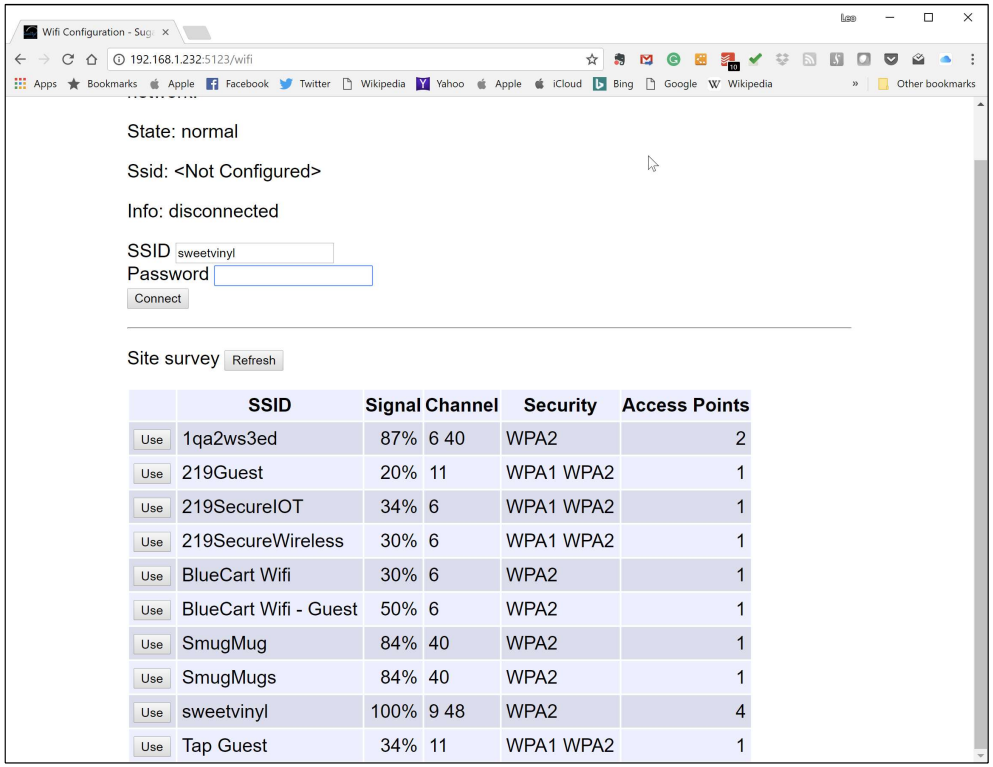
(1) Click on 'Settings'



(2) Display choices include showing the routing mode, show System Information, and show SweetVinyl splash screen. (**Note**: System Information such as **IP Address** of the SugarCube and **Software Version** can be obtained from the front panel by pressings and holding the '**Bypass**' button for about **five seconds**.)

4.7 Controlling the SugarCube via **WiFi**



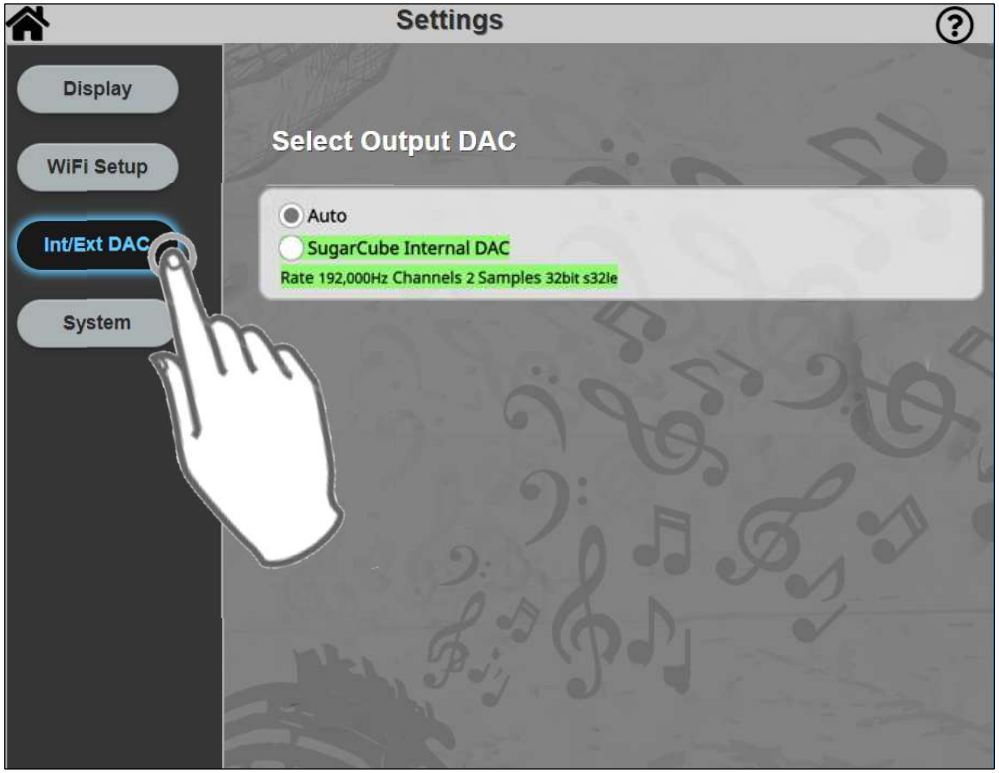


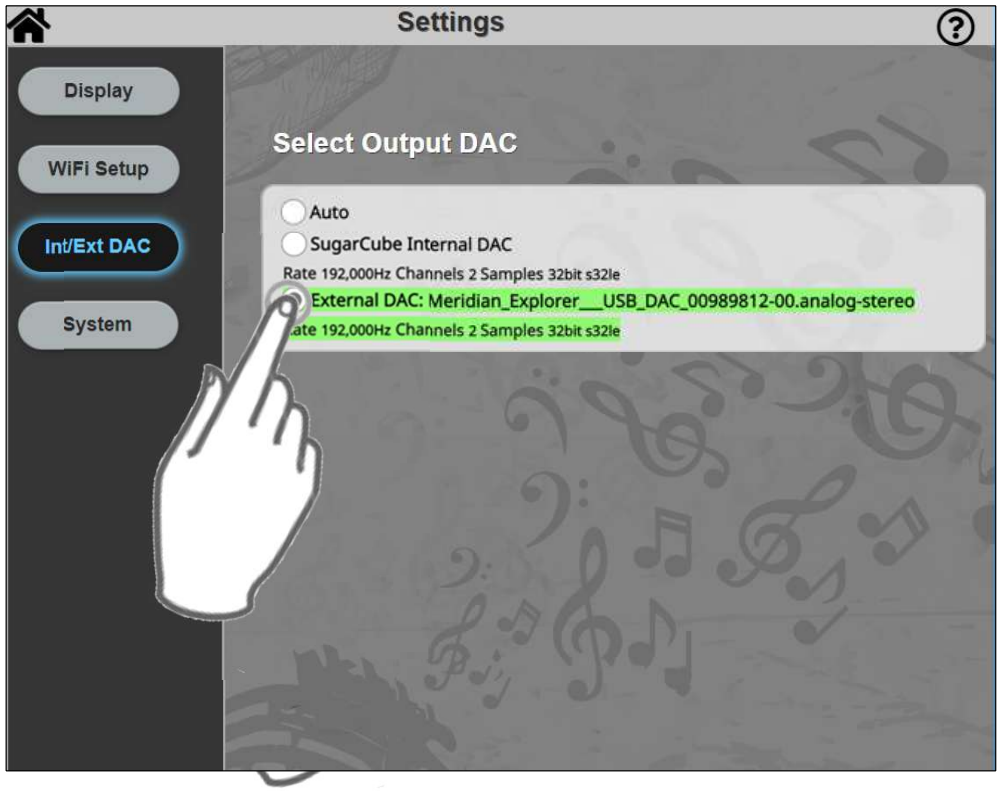
(1) Click on ‘WiFi Setup’

(2) When the Wifi dongle is in the rear USB port available networks will appear and you can select the SSID and enter your password.

4.8 Choosing an **External DAC** Output

Selecting an external DAC plugged into the SugarCube

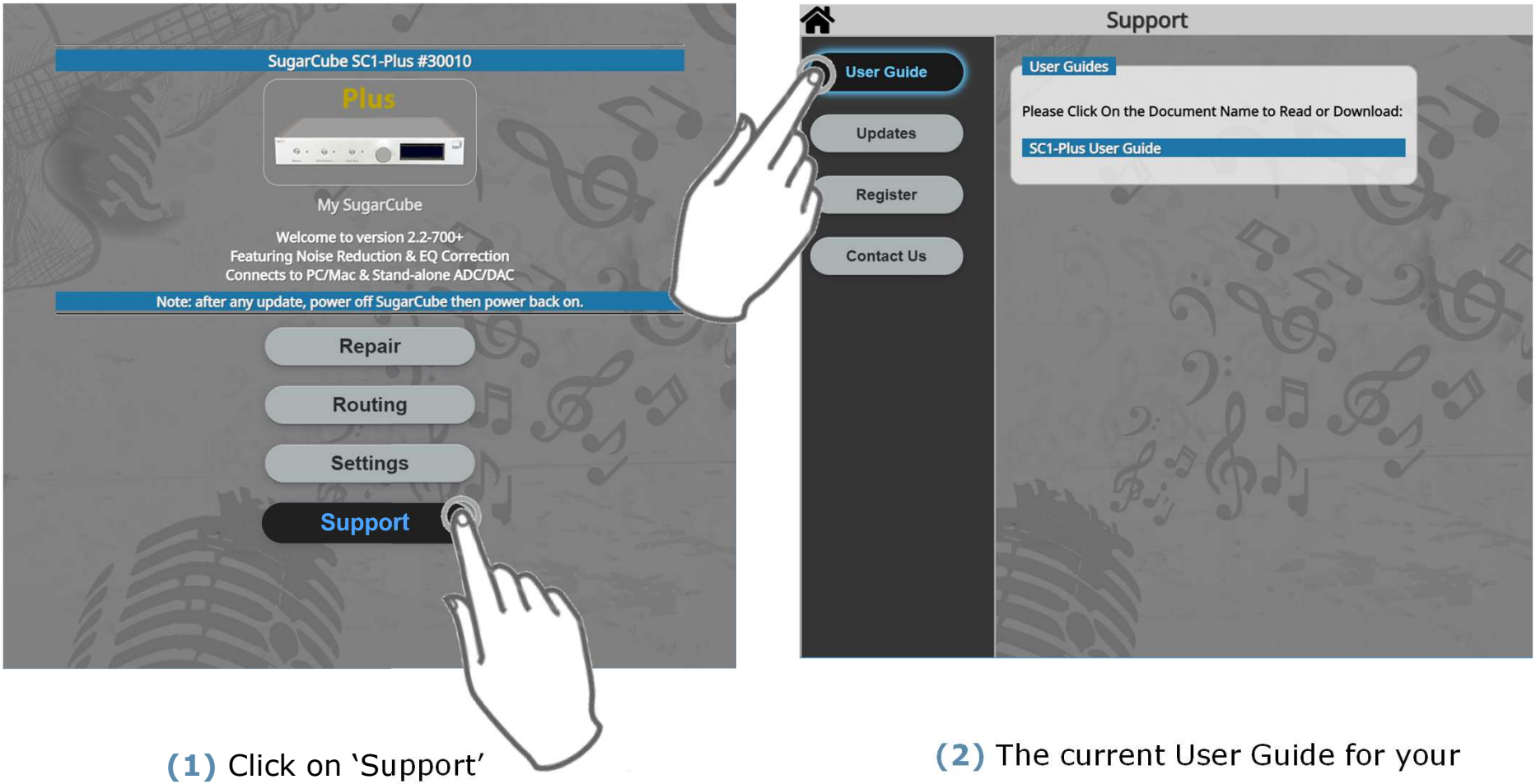




(1) Click on 'Int/Ext DAC.'

(2) When a USB DAC is inserted in one of the rear USB ports, it will appear here and can be selected instead of the internal DAC.

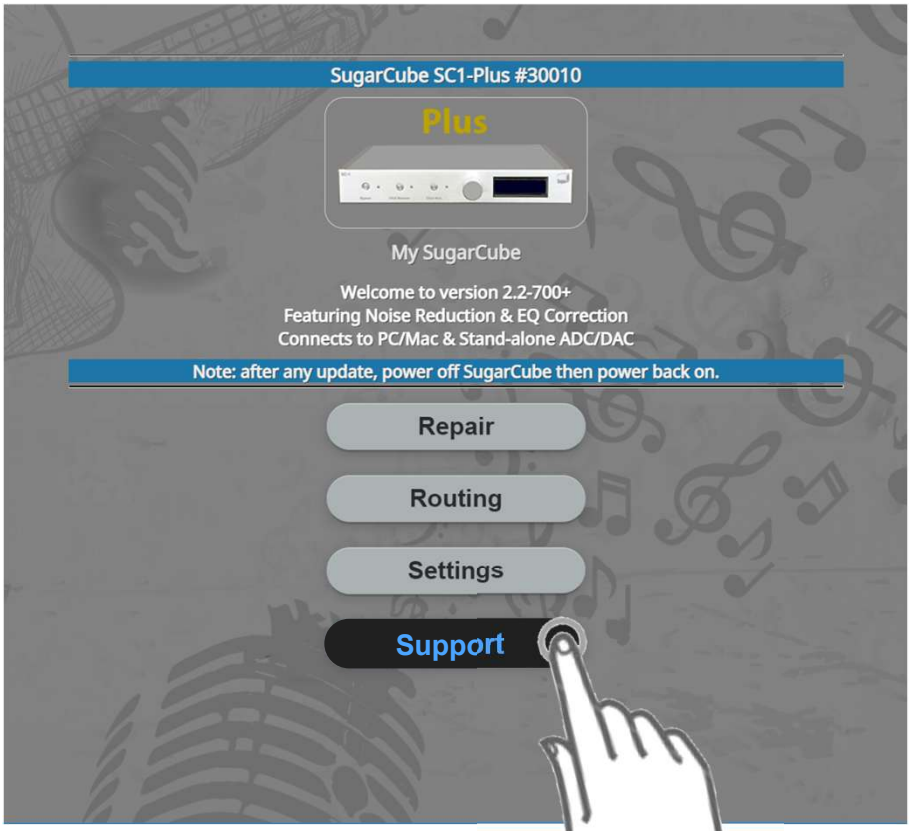
4.9 **Support** Menu & **User Guide**



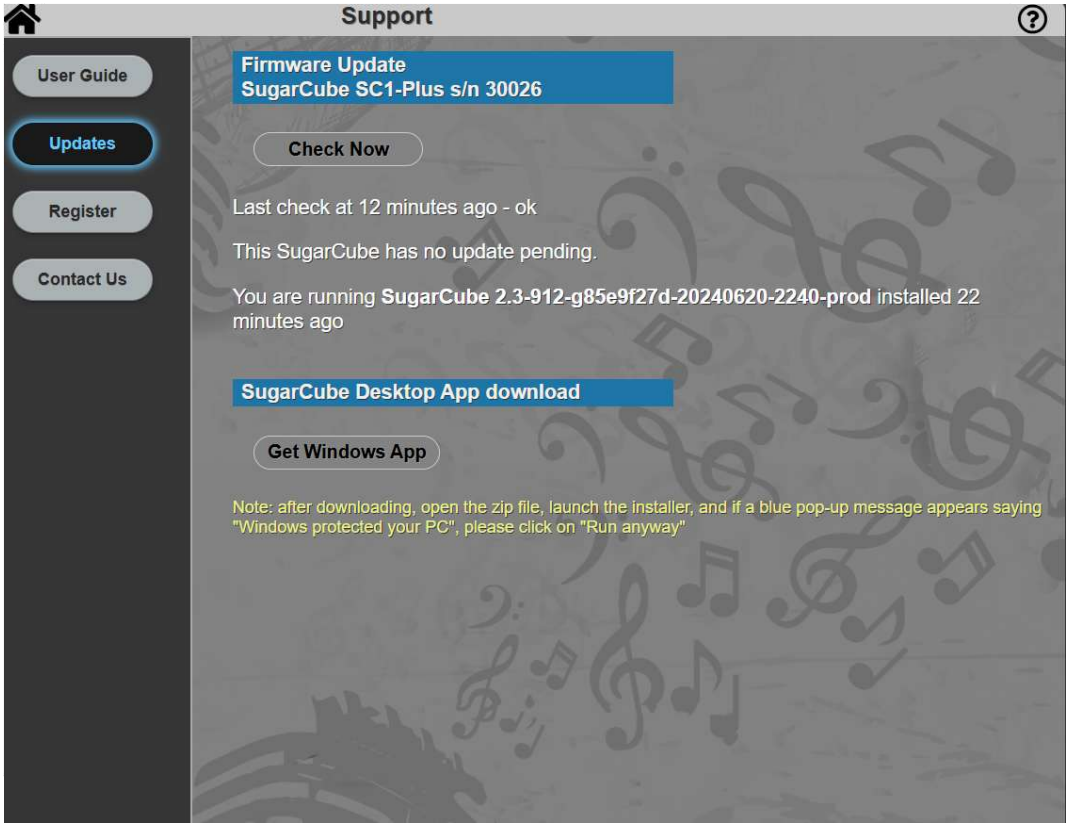
(1) Click on 'Support'

(2) The current User Guide for your model is available for download and/or viewing in the browser here.

4.10 Checking for **Updates**



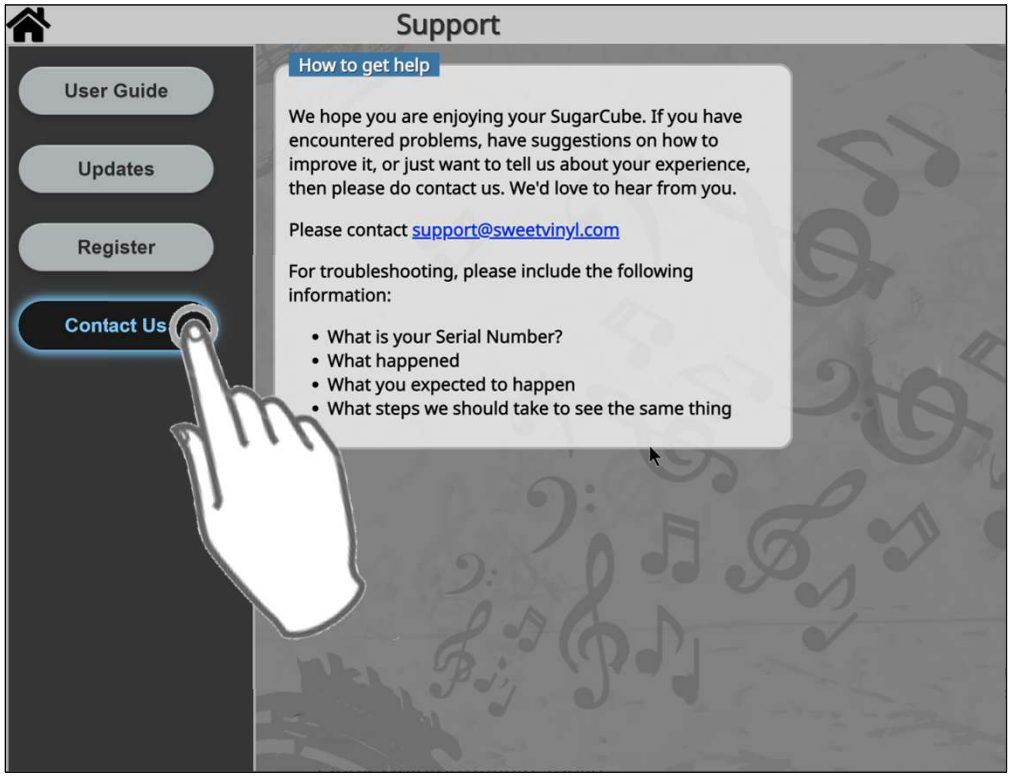
(1) Click on 'Support.'



(2) Current SugarCube software version is displayed and udpates can be checked. If updates are waiting can be updated here. Also shows last update and if an update is coming.

4.11 **Register** and Email Info





(1) Click on 'Register' and you can enter your information.

(2) 'Contact Us' will taker you to a screen where you can follow instructions to send an email to support@sweetvinyl.com.

Section 5

Connecting a Computer over USB

MacOS, Windows, or Music Server



5.1 **Connecting** a Computer over **USB**

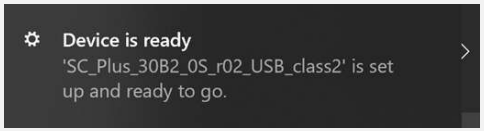


The SC1-Plus can be connected to a computer via the **USB-B** port. Plug one end into the USB port on the computer and the other to the **USB-B** port on the SugarCube. When connected, the computer will be able to control the **ADC** and **DAC** functions on the SugarCube. The computer can record the output of the **ADC** with any software available for recording or play digital files directly through the SugarCube.

5.1.1 Configuring Windows to control the SugarCube

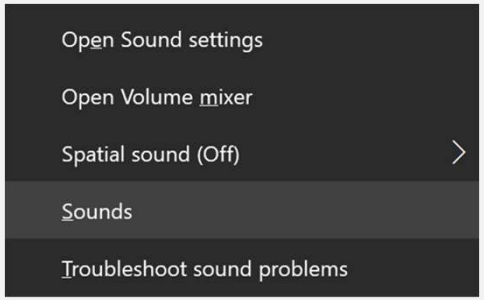
1

After connecting your Windows PC, the SugarCube should configure itself.



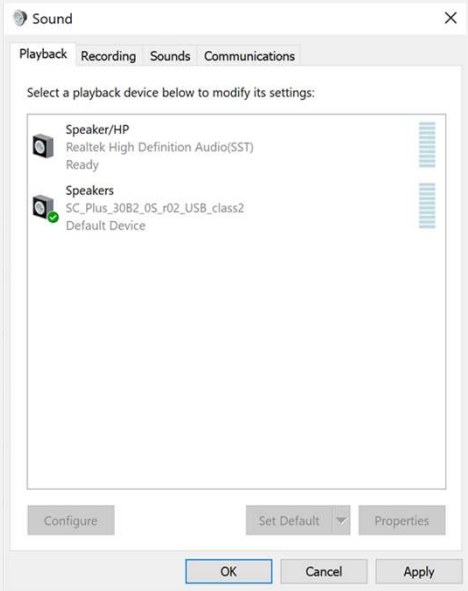
2

Drivers are built in and you should be able to see and configure the SugarCube as a device under Sounds by right clicking on the speaker Icon on the task bar.



3

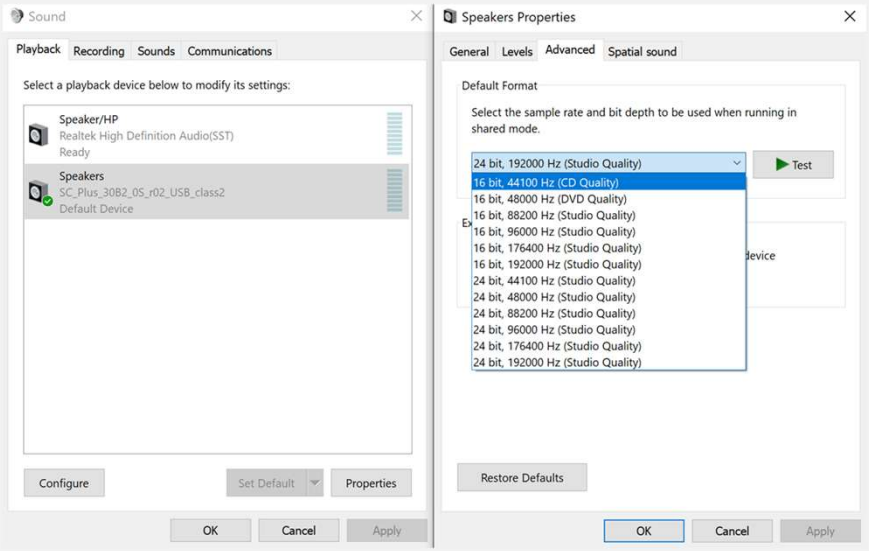
A Window like this will open:



5.1.2 **Configuring** Windows to control the **SugarCube**

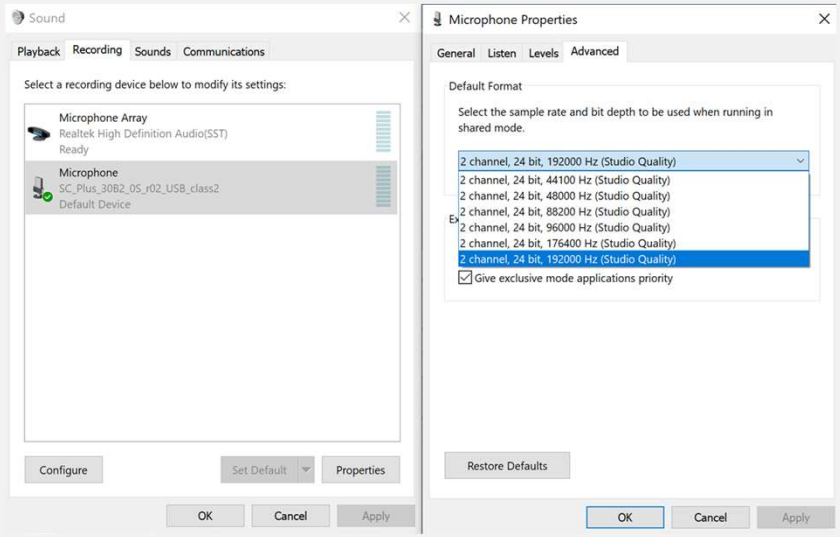
4

As a Playback device you will be able to configure the SugarCube DAC as a Speaker – by selecting the SC Plus you can than select Properties and the Advanced tab and select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.



5

As a Recording device the SugarCube ADC will show up as a Microphone by selecting the SC Plus you can than select Properties and the Advanced tab and select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.

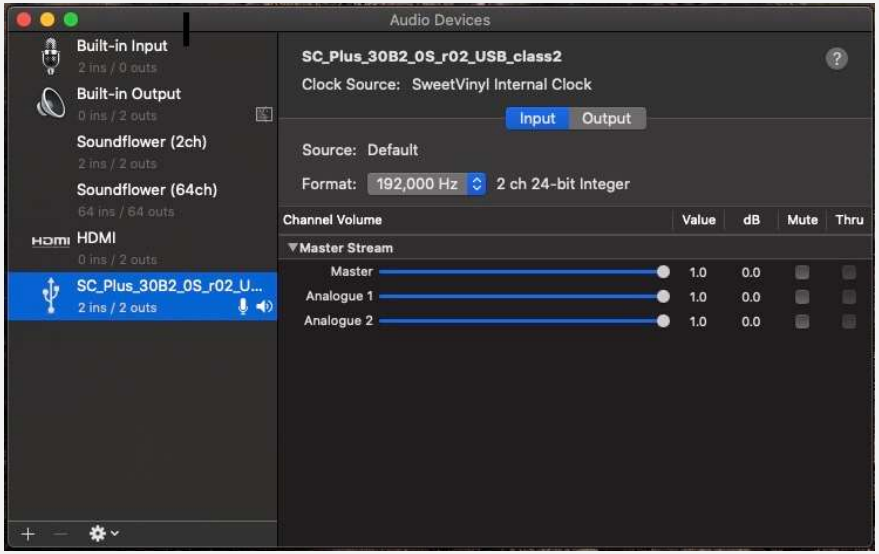


5.1.3 Configuring MacOS to control the SugarCube

1. After connecting your MacOS computer, the SugarCube should configure itself

2. Drivers are built in and you should be able to see and configure the SugarCube as a device under “Audio Midi Setup” by searching for this or by going to the "Utilities" folder in "Applications"

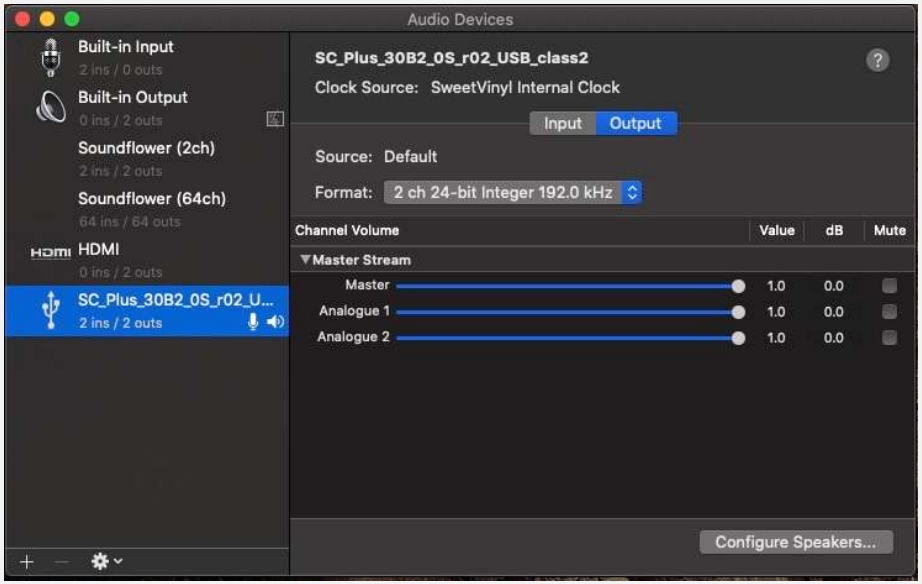
3. A Window like this will open and you will want to further select the “Audio Setup” section:



5.1.4 Configuring MacOS to control the SugarCube

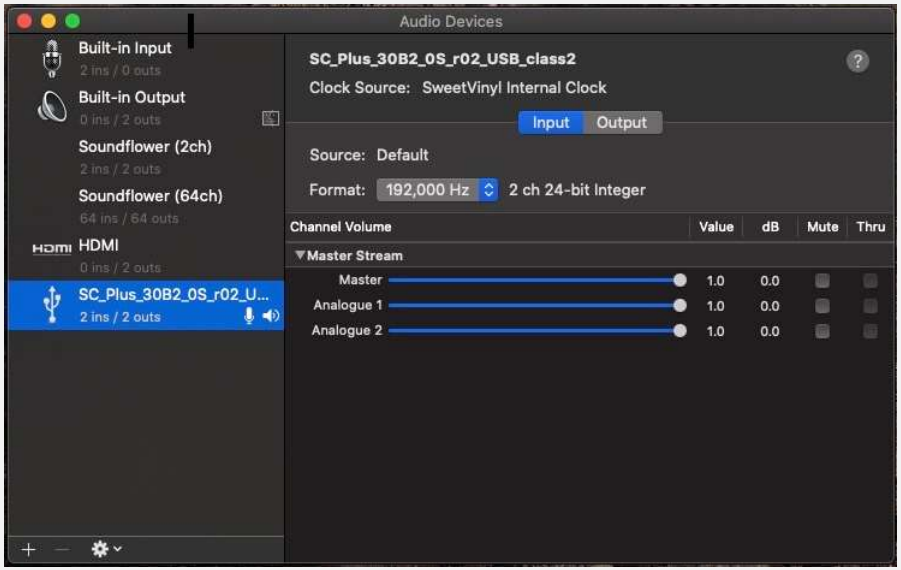
4

As an Output device you will be able to configure the SugarCube DAC– by selecting the SC Plus you can the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.

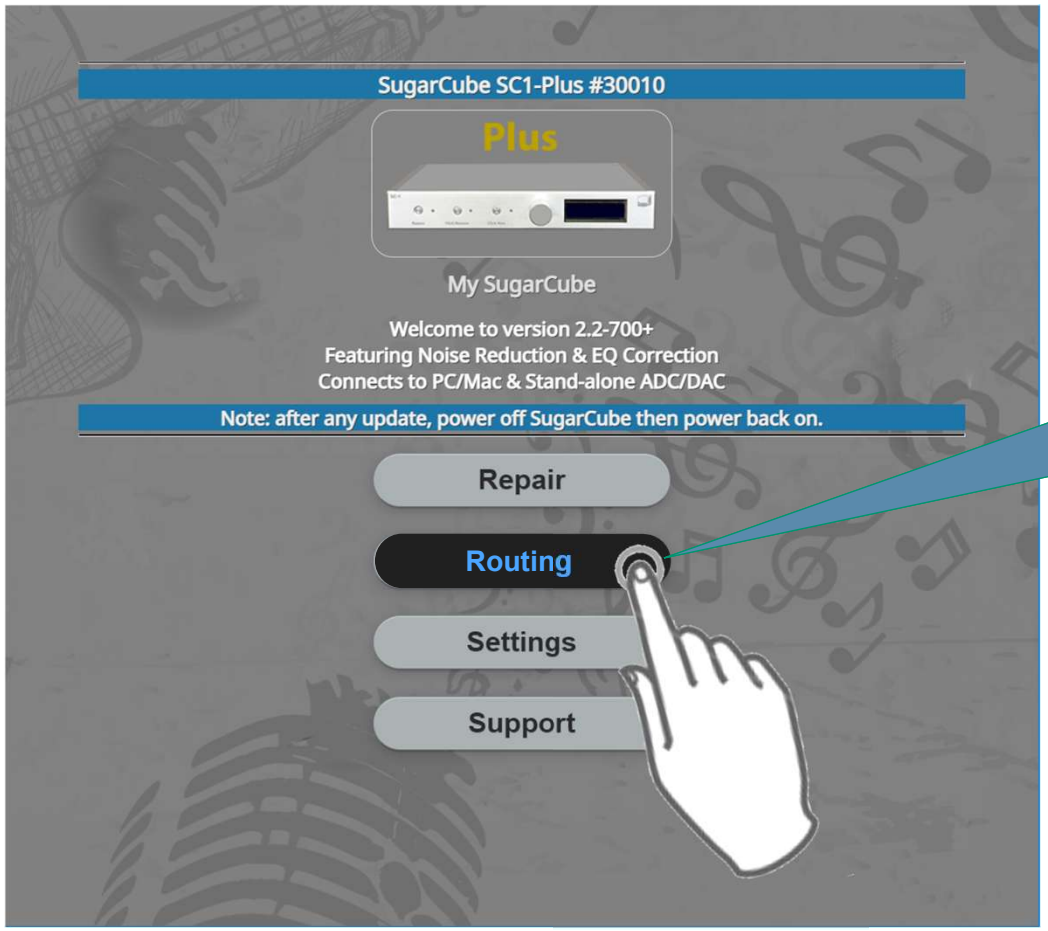


5

As an Input device the SugarCube ADC will appear and by selecting the SC Plus you can select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.



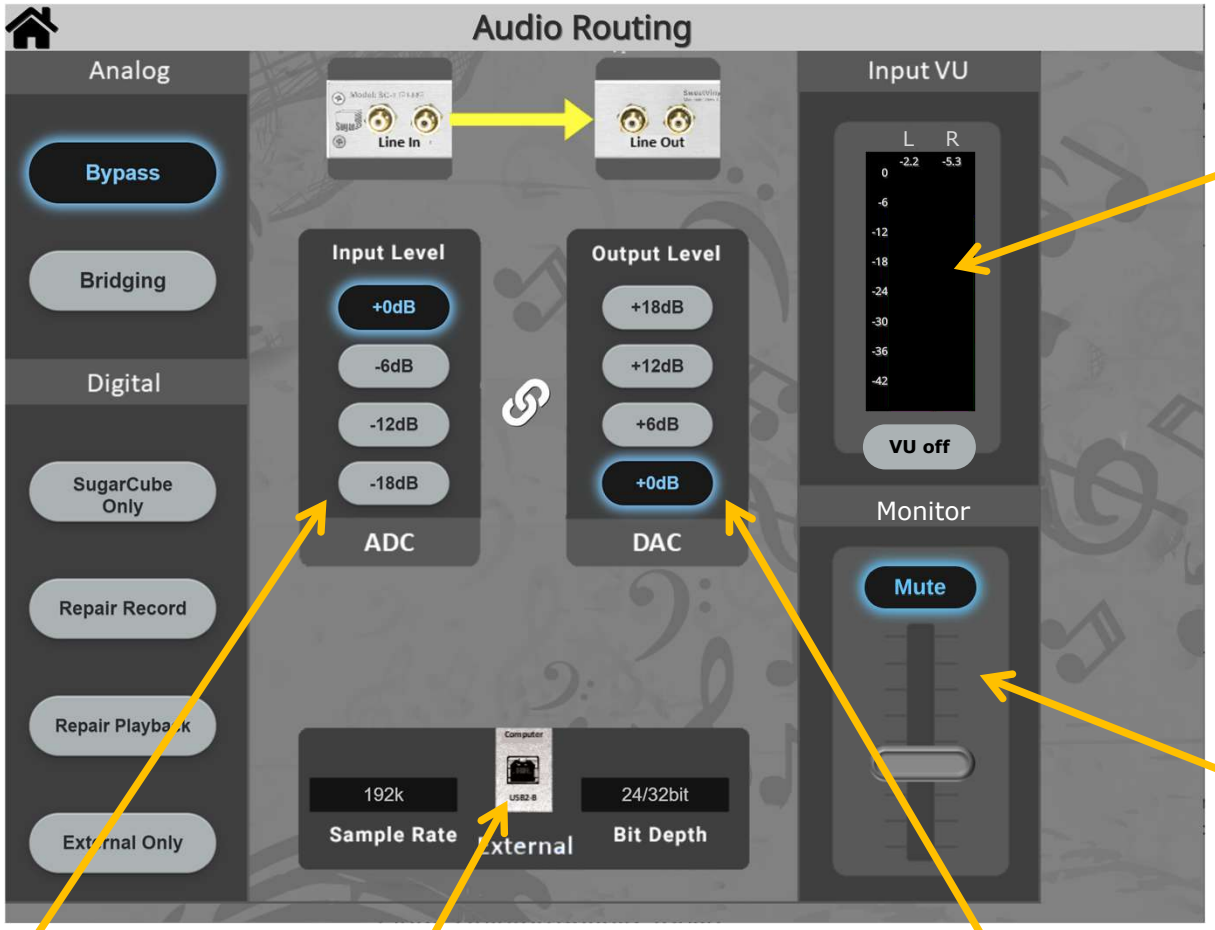
5.2 Audio Routing Menu



Select the **Routing Menu** to manage the analog and digital signal paths.

To efficiently work with an the SC1-Plus and a computer connected over USB you will want to open the **Routing Menu** page. The next pages show how the SugarCube, and an external computer, work together to provide the enhanced functionality.

5.3 Audio Routing Menu — Control Layout



Analog Routing Modes

Digital Routing Modes

VU Meter shows level of input as applied to the ADC

Level of audio sent to browser as a convenience monitor of input signal

Line-In level setting controls the level of **input audio** to the **ADC**

External computer or music server connects to SugarCube via **USB2**.

The **SC1-Plus** appears as an **ADC/DAC** to the computer

Line-Out level setting controls the level of **audio gain after** the **DAC**

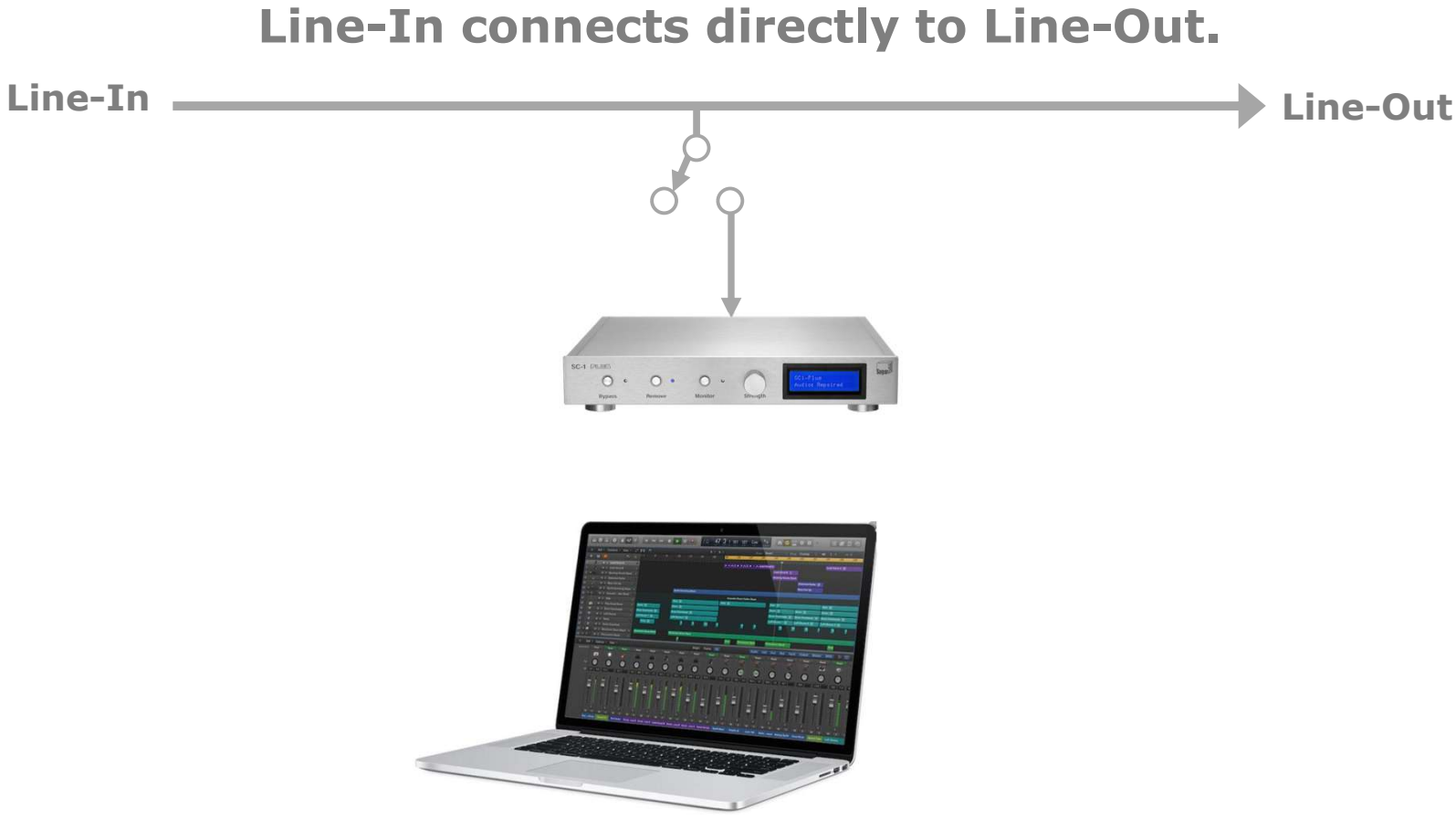
5.4 **Audio Routing** Menu — **Bypass** Mode



When **Bypass** is selected, **Line-In** connects directly to **Line-Out** thru an audiophile-grade mechanical relay.

In **Bypass** mode, **Line In** connects to **Line Out** via a mechanical relay; this can be selected on this menu or on the front panel. When in **Bypass** no signal is processed by the SugarCube nor by a connected computer.

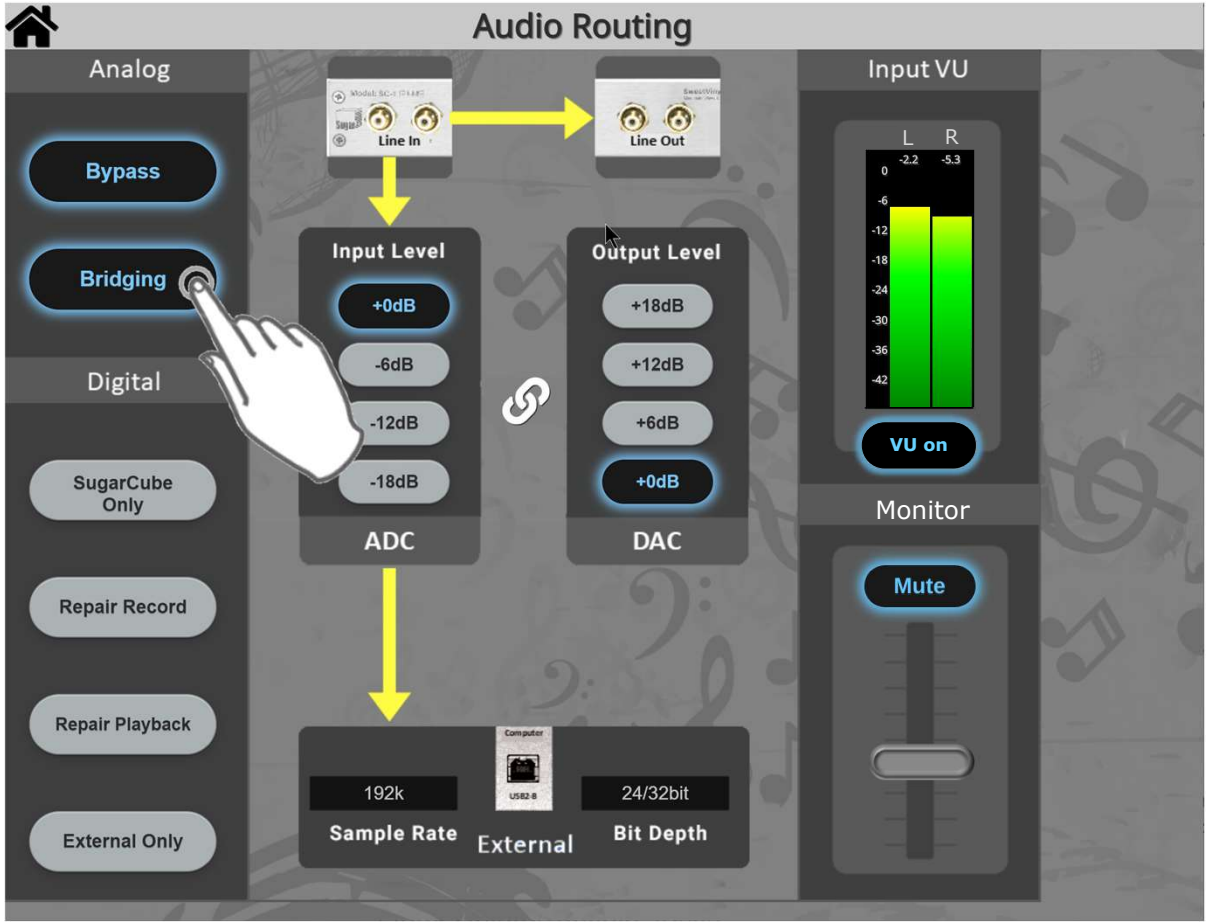
5.5 Logical Flow — **Bypass** Mode



Bypass | Bridging | SugarCube | Repair Record | Repair Playback | External

In Bypass mode **Line-In** is connected to **Line-Out** via a mechanical relay, this can be selected on the menu or on the front panel. When in **Bypass**, no signal is processed by the SugarCube or a connected computer.

5.6 **Audio Routing** Menu — **Bridging** Mode



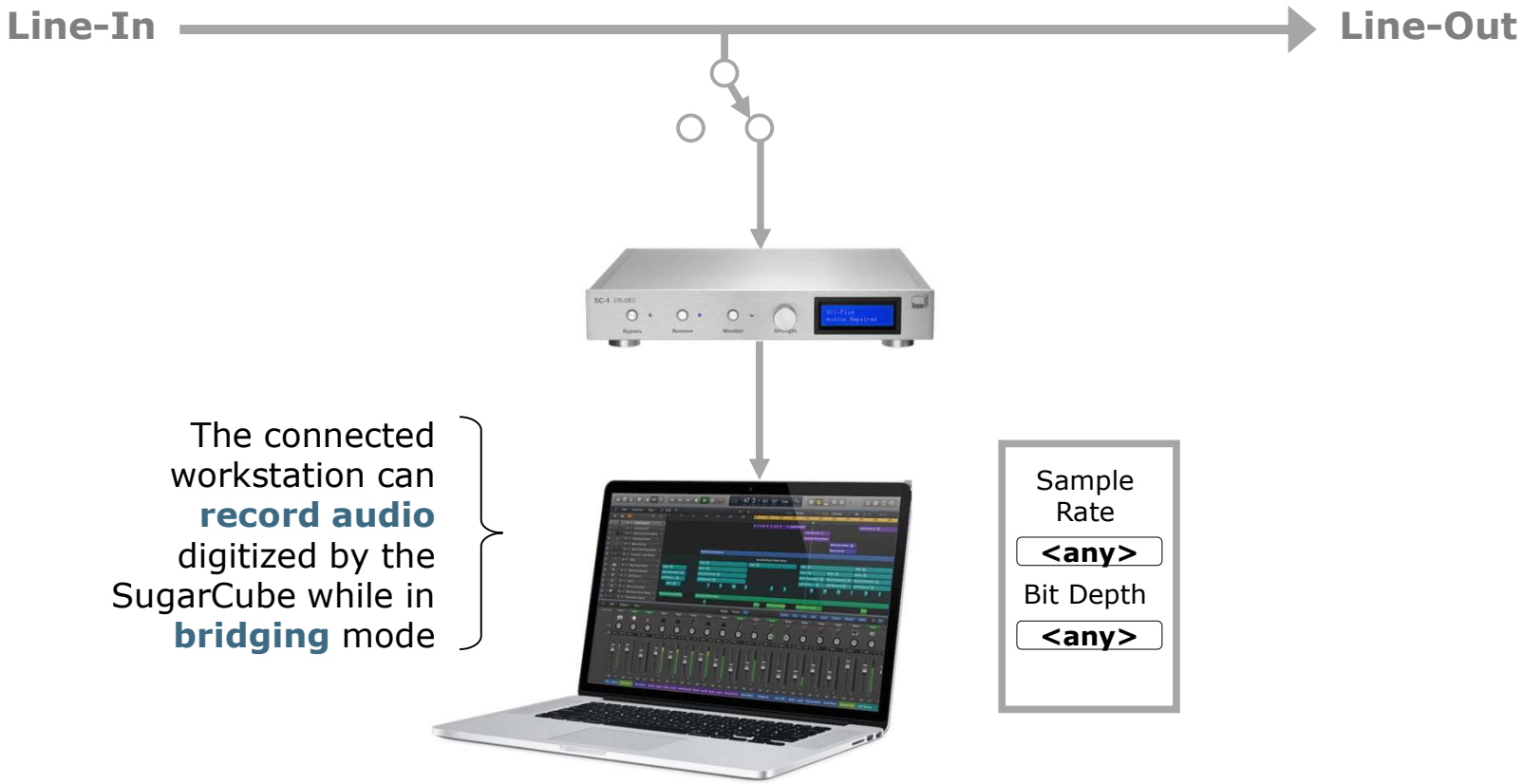
There is no audible affect as **Line-In** passes directly to **Line-Out**.

Bridging mode passes **Line-In** to the Analog-to-Digital Converter (**ADC**) which passes its digital output signal to the attached computer over **USB**, output only.

The **ADC** operates at 192k/24bits by **default** but the external computer can change to any standard sample rate and bit depth.

5.7 Logical Flow — **Analog Bridging** Mode

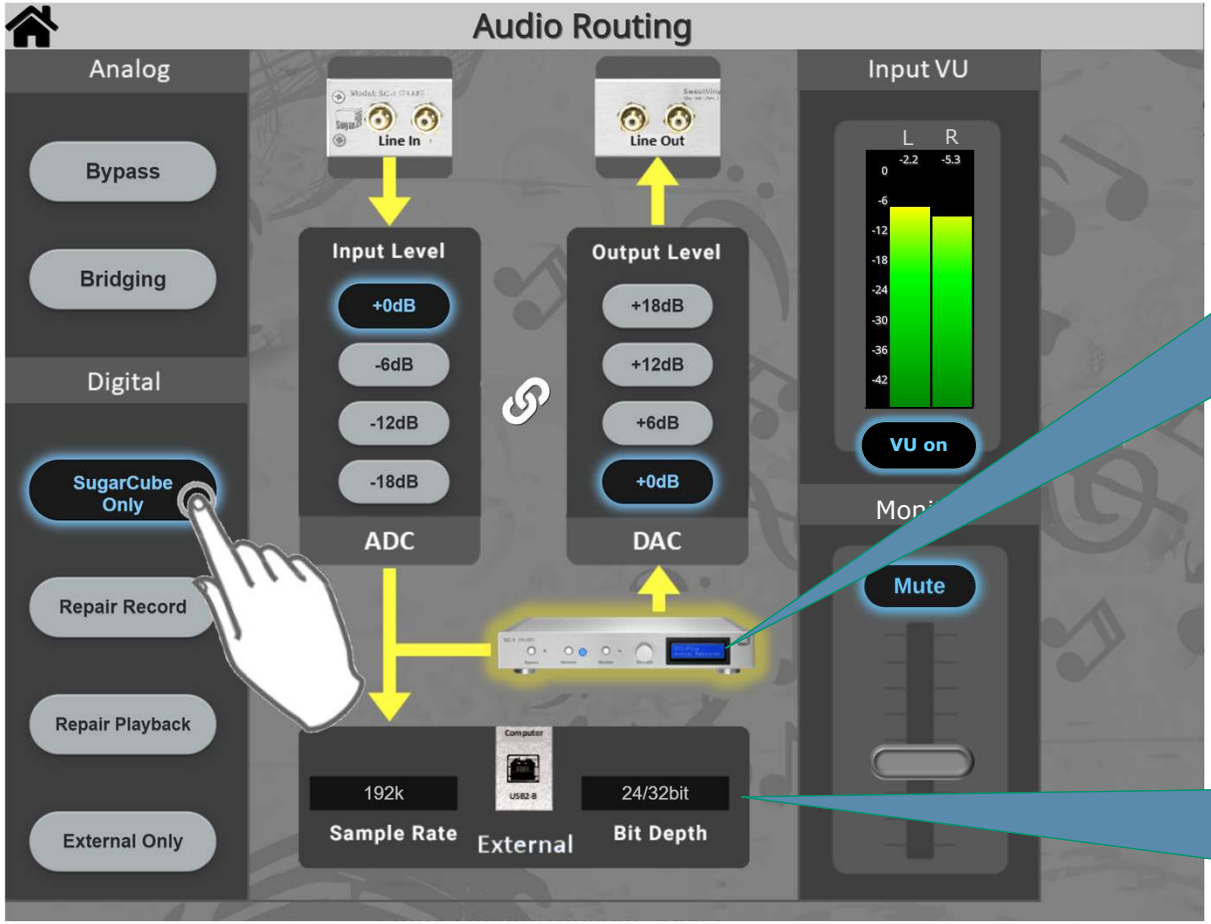
Line-In connects directly to Line-Out and SugarCube bridges the signal (with high impedance.)



Bypass | **Bridging** | SugarCube | Repair Record | Repair Playback | External

In **Bridging Mode**, **Line-In** is connected to **Line-Out** as in Bypass mode, in addition the signal is passed to the attached computer via the SugarCube's **ADC**.

5.8 Audio Routing Menu — SugarCube Only

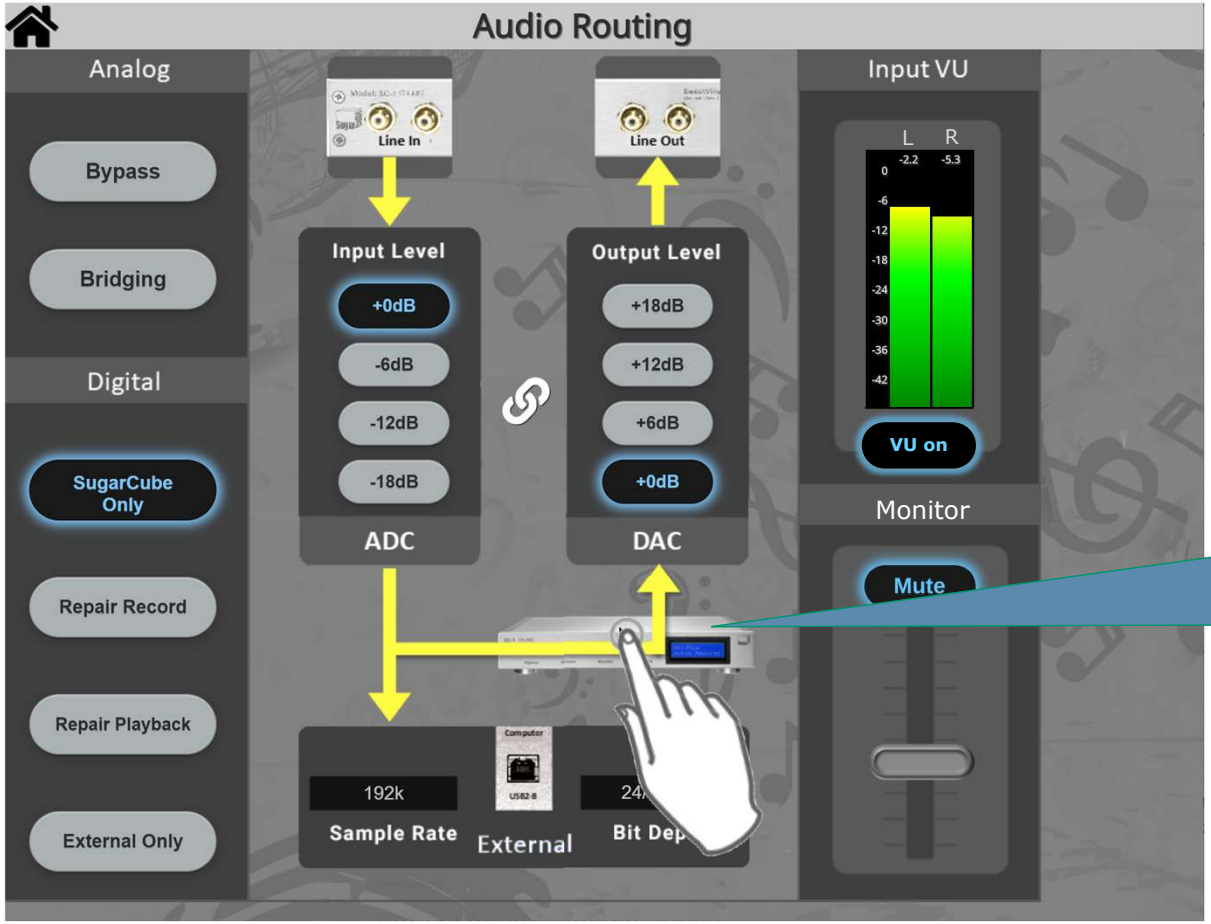


Audio Passes from **Line-In** through the **ADC** then into the **SugarCube** then into the **DAC** to **Line-Out**

Raw digital audio from the **ADC** output is also sent over **USB** to the attached computer at 192K/24bit

In **SugarCube-Only** mode, audio is processed by the SugarCube as standard on all Sugar Cubes. In addition, the **Un-Repaired Digital** signal is passed to the attached computer via the **ADC**.

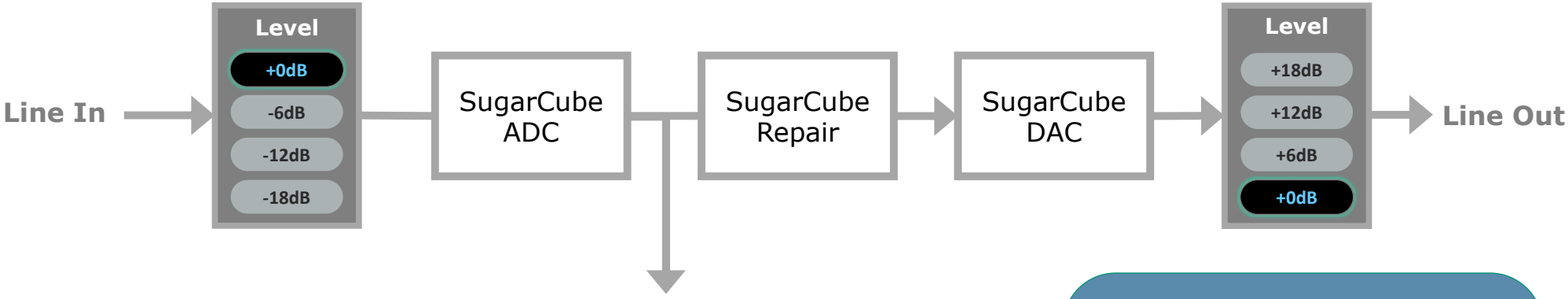
5.9 Audio Routing Menu — SugarCube Only



Click on the SugarCube **icon** to engage/dis-engage repair functions

5.10 Logical Flow — SugarCube Only

Normal SugarCube operations but with digital signal from ADC tapped and send out to workstation.





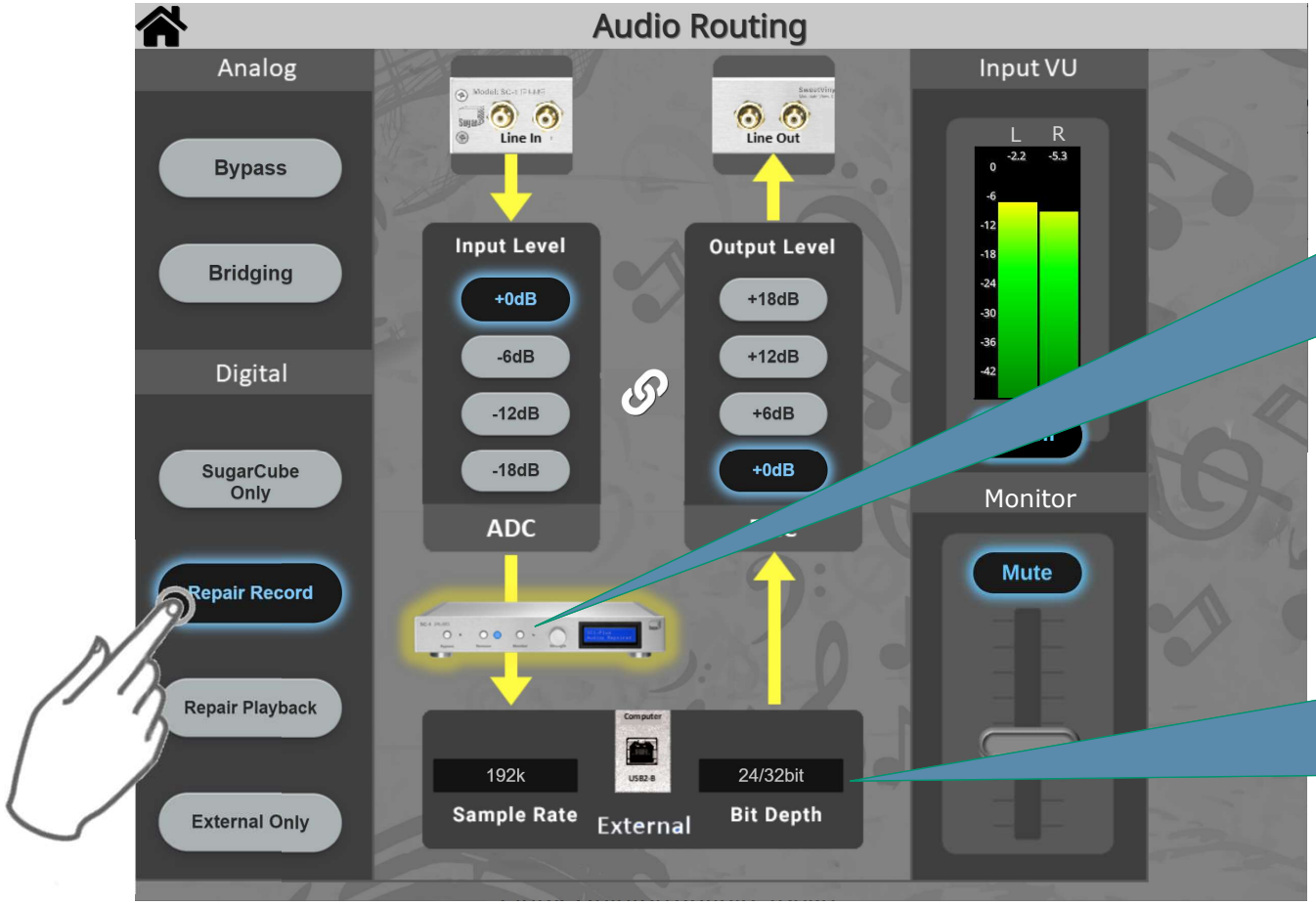
Sample Rate
192k
Bit Depth
24bit

Please note that when the SugarCube is "in the loop", the only available output sample rate is **192k/24bit**

Bypass | Bridging | **SugarCube** | Repair Record | Repair Playback | External

In **SugarCube-Only** mode, **Line-In** is processed by the SugarCube and audio is repaired and sent to **Line Out** as standard on all SugarCubes. In addition, the repaired signal is passed to the attached computer via the **ADC** and the computer can output digital audio thru the **DAC**.

5.11 Audio Routing — Repair Record



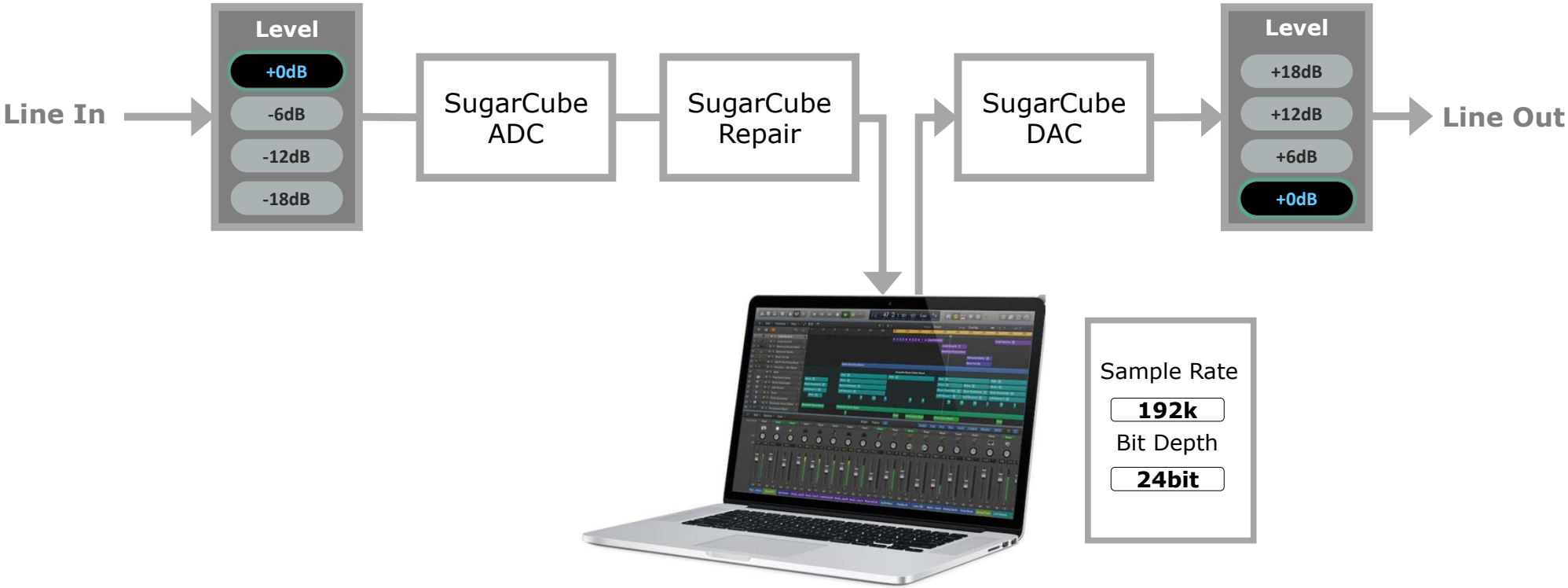
'**Repair Record**' inserts the SugarCube repair processing **after** the **ADC** to repair audio before outputting to the external device

If the connected computer operates at **192k/24bit**, the system will be able to utilize SugarCube repair functions

Note: Please **press again** the button of **any mode** of operation (even though it is already selected) if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

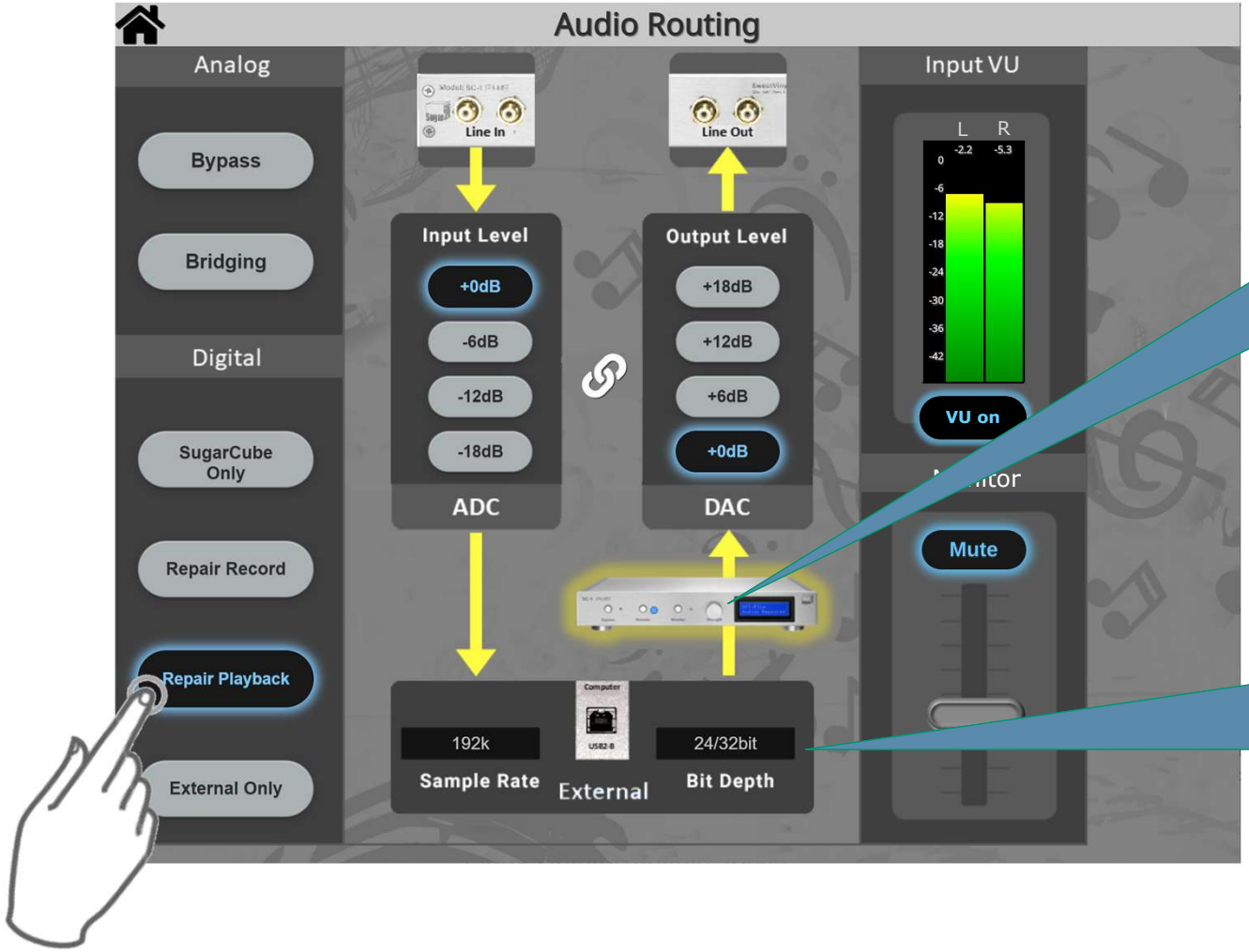
5.12 Logical Flow — **Repair Record**

SugarCube repairs audio before sending out to workstation.
Workstation directly drives the DAC.



Bypass | Bridging | SugarCube | **Repair Record** | Repair Playback | External

5.13 Audio Routing — Repair Playback



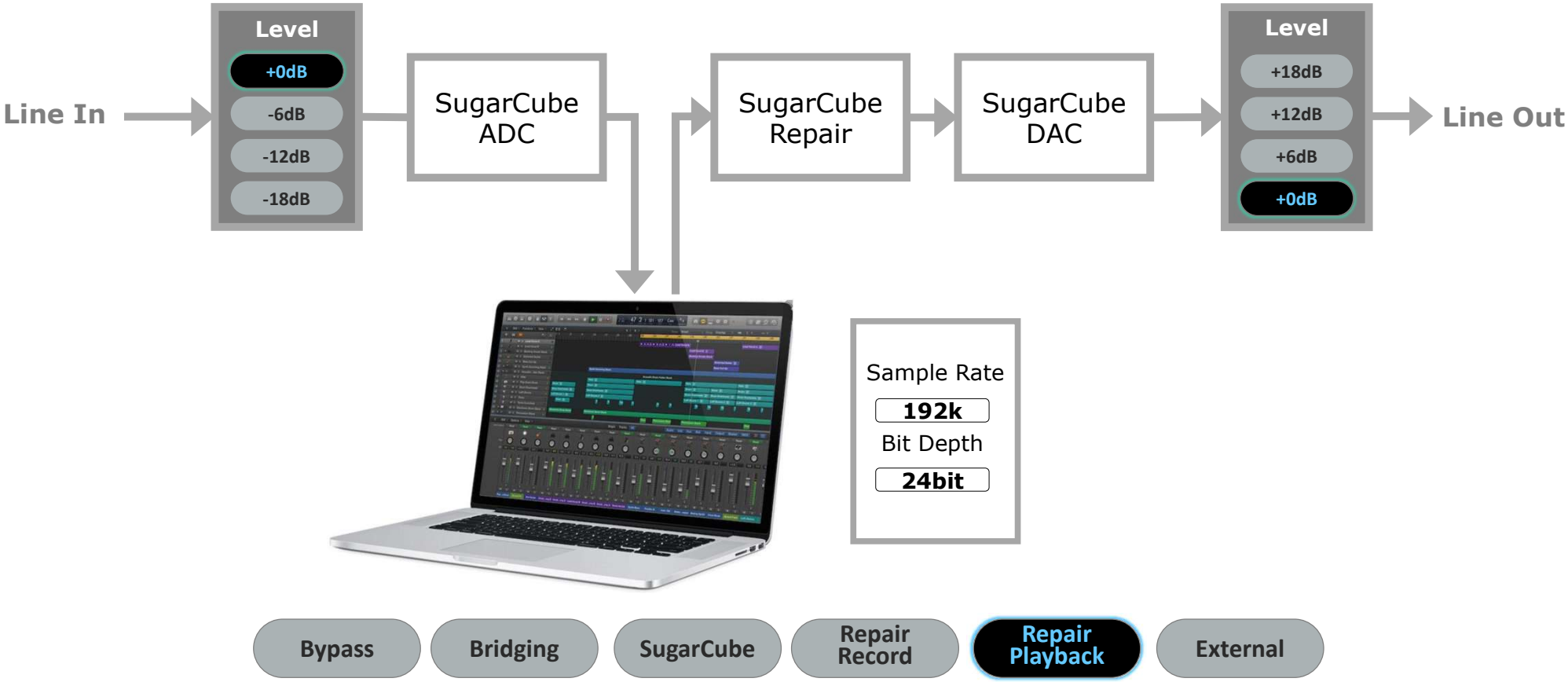
'**Repair Playback**' inserts the SugarCube repair processing **before** the **DAC** to fix incoming digital audio from the external device

If the connected computer plays a digital recording at 192k/24bit, the system will be able to engage SugarCube repair functions

The **Repair Playback** mode is particularly useful allowing you to play any digital file from your library to remove clicks & pops and/or remove tape hiss from previous recordings from any source from the past.

5.14 Logical Flow — **Repair Playback**

Digitized audio goes directly to workstation. Workstation output feeds SugarCube for repair before DAC.



Repair Playback is our **recommended mode** for operation. All of your files will be **recorded** "in the raw" with no SugarCube processing applied. Then, on **playback**, you can adjust click & pop removal as well as apply noise reduction (if needed) on a recording-by-recording basis or use no processing at all if desired.

5.15 Audio Routing Menu — External Only

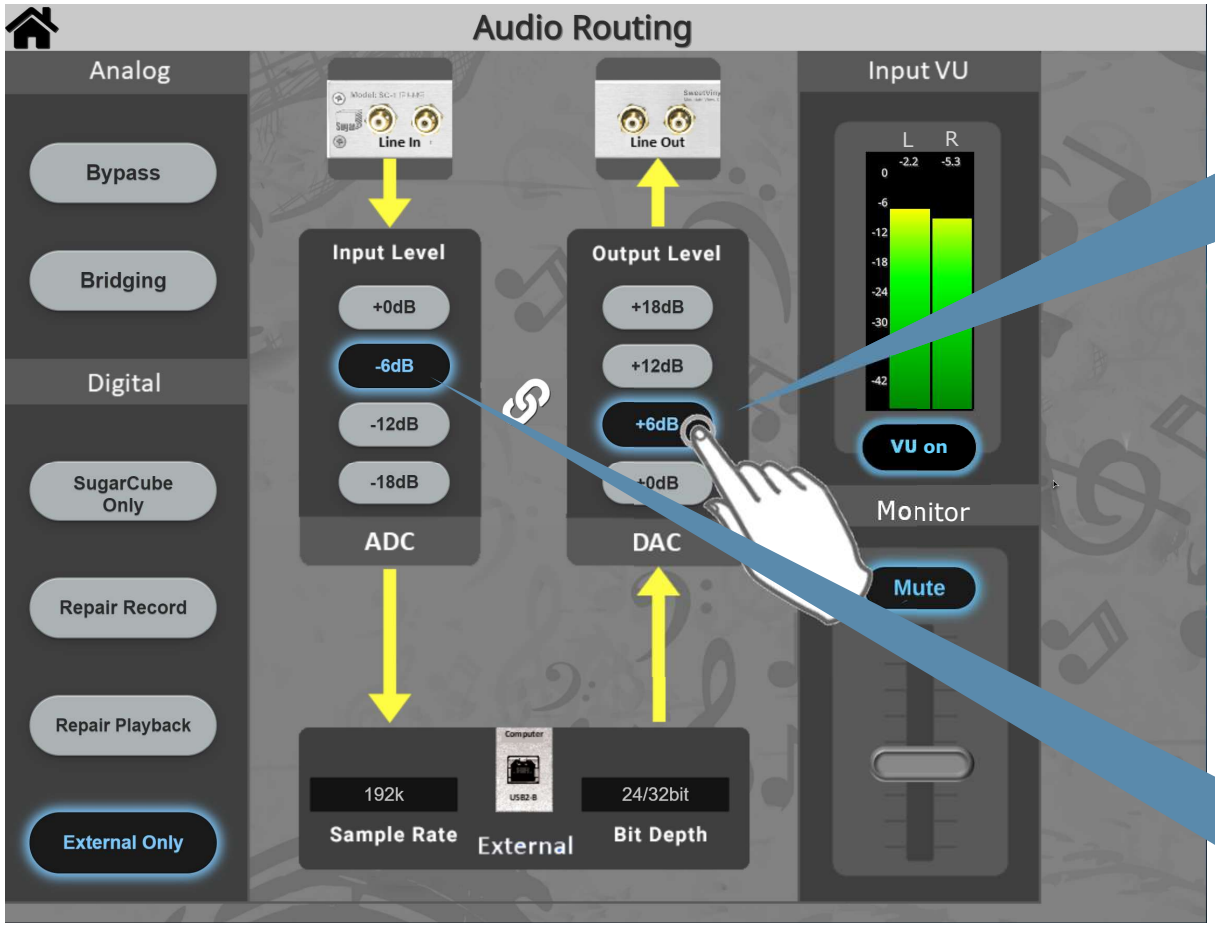


External-Only mode allows an external device to use the **ADC** and **DAC** at **any sample rate.**

In **External Only** mode, the **SC1-Plus** acts like **any other** studio-quality **ADC/DAC** pair and will function at any standard sample rate and bit depth from 192k down to 44.1k and at 16/24/32 bit sample sizes.

Note: Please **press again** the button of **the current mode** of operation if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

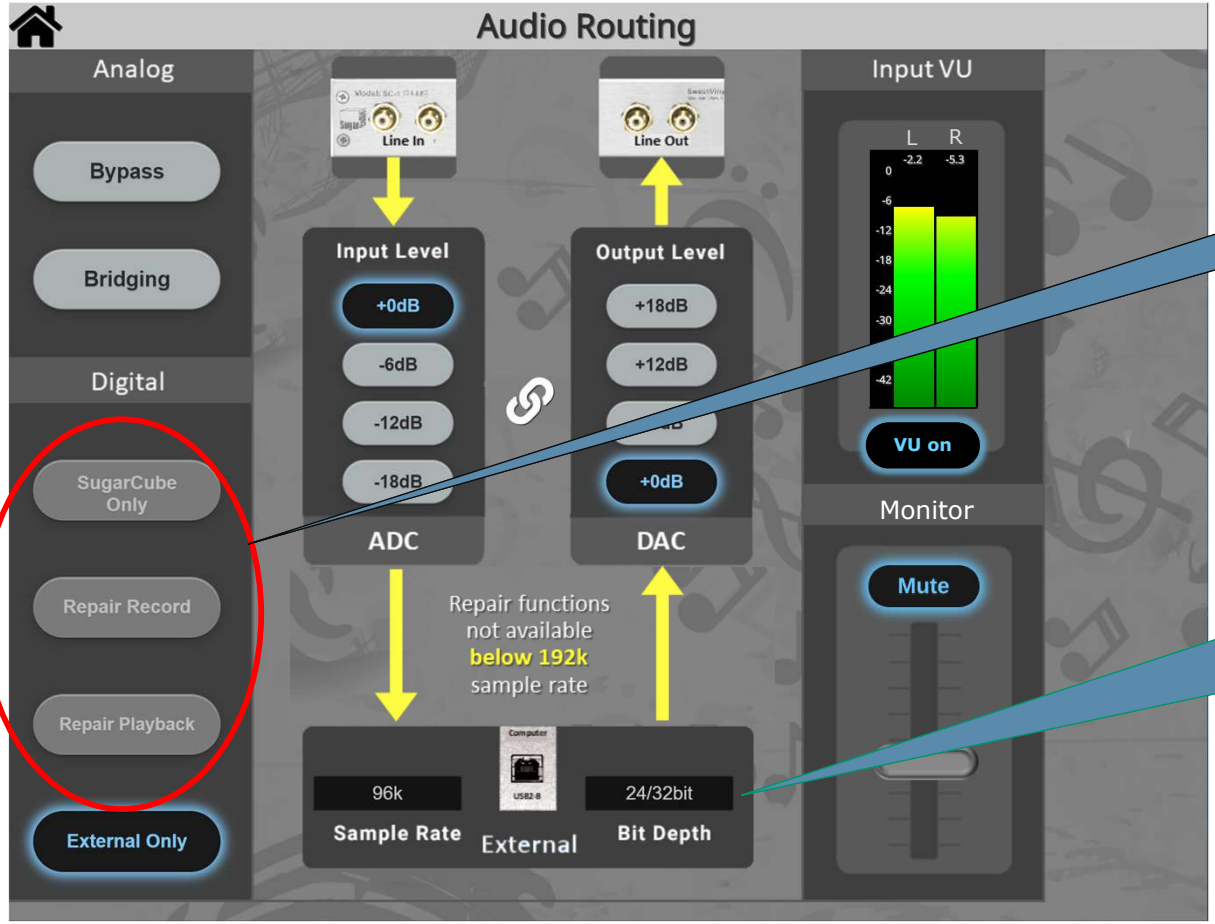
5.16 Audio Routing Menu — External Only



Output Level can be changed during playback to any of the four available levels.

However, input level will change in the **opposite** direction.

5.17 Audio Routing — External Device **Not** Set to 192k/24bit



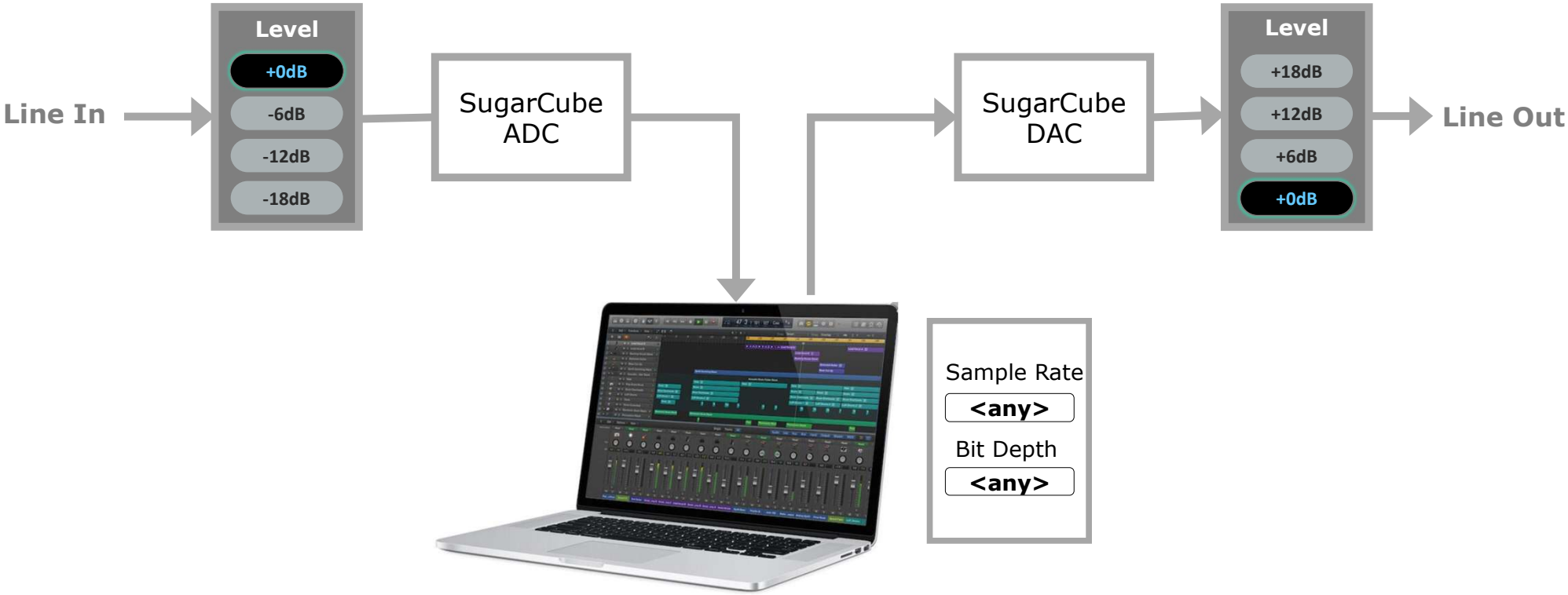
These repair modes are **not available** below **192k/24bit**

The **SugarCube** will operate as an excellent **ADC/DAC** at any sample rate or bit depth just **no repair functions** are available **below 192k/24bit**.

In **External Only** mode, it is fine to operate at **any sample rate and bit depth below 192k/24bit**, but SugarCube **repair functions** do not operate **below 192k/24bit**.

5.18 Logical Flow — External-Only

SugarCube repair functions are out of the loop.
Workstation uses ADC & DAC directly



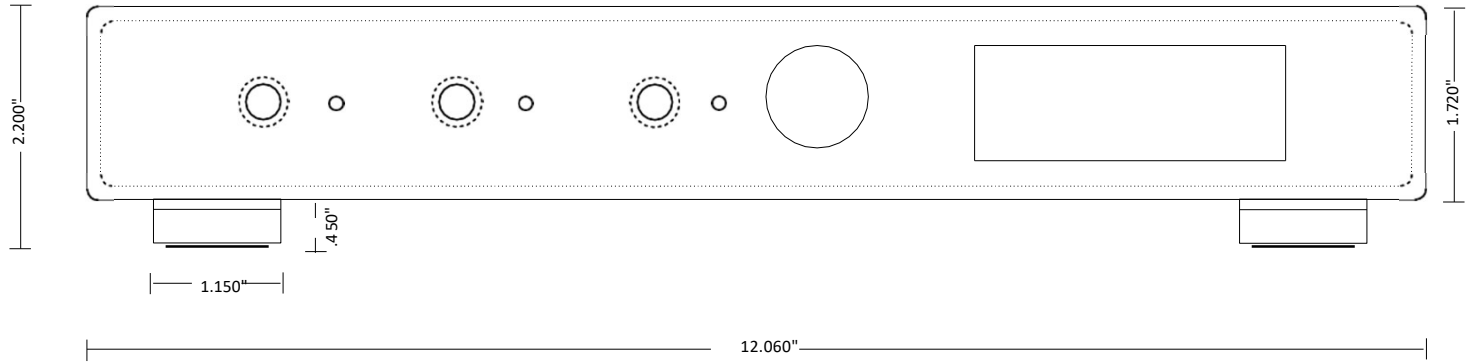
In **External-Only** mode, the SugarCube will operate at **any sample rate and bit depth** from 192k/24bit down to 44.1/16bit, just like any other studio-quality ADC/DAC pair.

Technical Specs

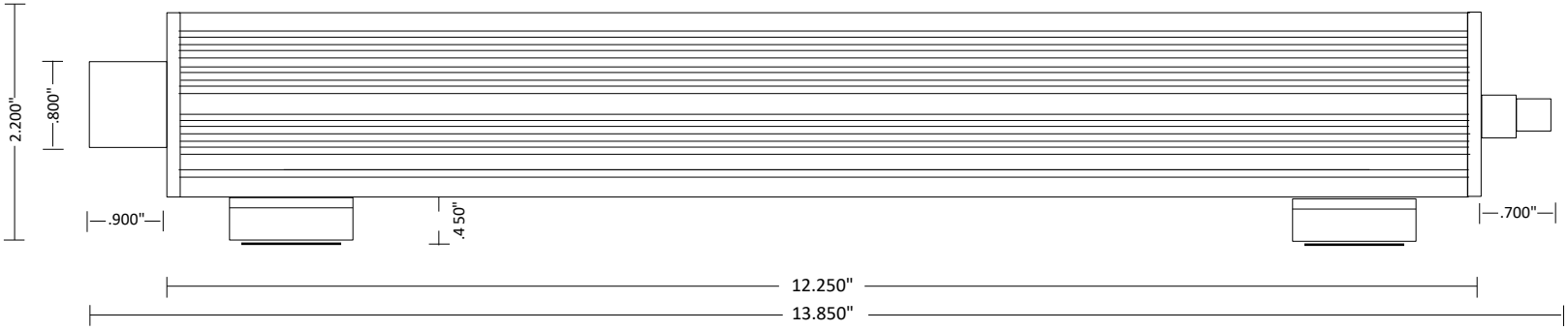


SugarCube SC1-Plus Mechanical:

Front View



Side View



Technical Specifications

- Line In maximum level = +22.7 dBu

  (approx. 10.6V rms or 30.0V p-p)
- Line Out level = Line In level
- THD+N = 0.0031% (A-weighted)
- Frequency Response = 25Hz to 60kHz +/-0.1dB
- Frequency Response = 15Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 11 watts

General Info

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: support@sweetvinyl.com

Trademarks

SweetVinyl™ and SugarCube™ are trademarks of SweetVinyl Inc.
Any non-SweetVinyl trademarks are used for reference purposes only.

Compliance

For the current EU Declaration of conformity, visit:
http://support.sweetvinyl.com/regulatory

For regulatory compliance information in North America, visit:
http://support.sweetvinyl.com/regulatory

SweetVinyl, Inc.

409 Tennant Ave, Suite 104
Morgan Hill, CA 95037



October 2026

FAQ

Some common questions & answers about using the SC-Plus series



Basic connectivity

- **Q: Line-In level setting**, why so many choices?
- **A:** You can select four steps each 6dB apart to accommodate low- to high-level output equipment. Whatever level you select for input is mirror the opposite way for output (i.e.--choosing -6dB line in results in +6dB applied to line out so line in always equals line out.)
- **Q: What is bridging?**
- **A:** This is an analog routing mode where line in goes directly to line out through the bypass relay just like in Bypass mode but is tapped at a high impedance (about 100k ohms) and passed through the internal analog-to-digital converter and then sent out to whatever is attached to the SC1-Plus digital interface. The attached computer can operate at any sample rate or bit depth in this mode.
- **Q: How do I connect through tape loops?**
- **A:** Tape Out goes to the SugarCube Line In and the SugarCube Line Out goes to Tape In. Sometimes the tape loop is labeled Aux or Rec but is the same thing.
- **Q: How do I connecting an external computer?**
- **A:** The SC1-Plus appears as a standard ADC/DAC to a connected PC, Mac, or music server. No special drivers are needed.
- **Q: My computer only goes to 48k, how to I fix that?**
- **A:** You can always use the SC1-Plus as an ADC/DAC at 48k but you cannot then use the SugarCube internal functions. You will need to disconnect the computer when you wish to use the SugarCube with your turntable.

Repair modes

- **Q:** when I use the **noise reduction**, do I have to **start at the lead in groove** or can I play any track on the record?
  - **A:** This is a yes and no answer: for the first time you play any record using noise reduction (NR), you must **start the noise reduction process then drop the needle** on the lead in groove.
  - **A:** Then, if you do not stop the process (that is leave it running), if you are playing, for example, a box set of records from the same album or a double album, the "fingerprint of the noise that the SugarCube created stays in memory and, as long as you don't turn off the SugarCube, you can drop the needle anywhere or play other side without having to drop the needle in the lead in groove. The NR process will continue to work perfectly.
  - **A:** Furthermore, you can extend the above concept if you have LPs from the same label and era (example: late-1950's Blue Note), the noise "fingerprint" that the SugarCube picked up on the first side of the first LP will work for all other LPs of your Blue Note collection when you use NR.
  - **A:** However, if you play an LP from say, EMI, that label used a different vinyl plastic formula in making their LPs so their (EMI's) noise "fingerprint" will be quite different and will require stopping the NR process and re-starting then drop the needle once again in the lead in groove.
  - **A:** Finally, you could use NR and only one time drop the needle for a "fingerprint" then never do it again, NR will still work but it will be far from optimal and there will be a drop in volume.
  - **A:** Bottom line: NR is a very powerful tool that can do an amazing job of lowering noise in recordings. It is well worth your time to start and stop NR when the LP is different.

Repair modes

- **Q:** What does **EQ** do?
  - **A:** LPs are made with a process that boosts the highs and diminishes the lows during the cutting of the master disk. This process was fundamental to record making all the way back to Thomas Edison's days in the early 1900's. However, until the mid to late 1950s, each manufacturer (record label) had their own process that used differing amounts of boost and cut. So, an RCA Victor record would not sound the same played on a non-RCA turntable.
  - **A:** More importantly, if you are playing 78s (using a 78-rpm phono cartridge) through your system, your phono pre-amp is almost surely using the RIAA equalization which became fully adopted in the early 1960s. RIAA has a much more drastic cut of high-frequencies and a correspondingly strong boost of the bass frequencies on playback than did any of the 78 recordings made prior to the emergence of the LP in the early 1950s. If you listen to a 78 even with a 78 needle, your phono pre-amp will cut much more treble and correspondingly boost much more bass than needed making the playback sound muddy and lacking in detail.
  - **A:** When you apply EQ on the repair page with the correct equalization choice from our menu, you 78 playback will sound remarkably cleaner and balances with detail that you did not know was there.

Audio routing and use with external devices

- **Q:** Why can’t my computer run Vinyl Studio at any sample rate I want?
- **A:** You can run at any sample rate/bit depth that you wish as long as you are operating your SC1-Plus either in the Bridging mode or External-Only mode. When you want to use the SugarCube processing with external recording, you need to operate at 192k/24bit to match our internal digital processing
- **Q:** Why would you want to remove clicks & pops during playback (Repair Playback mode) instead of during recording (Repair Record mode.)
- **A:** There are a reasons to operate your SC1-Plus with Repair Record as well as with Repair Playback. (1) for the Repair Playback mode, you can record batches of LPs without any processing and then apply click and pop removal and/or noise reduction on playback on a case-by-case basis just like might you do when you listen to an LP. If you were to use Repair Record, you would want to be sure you had all your settings optimal because once recorded, you cannot change anything. However, if you were making a lot of recordings to save, say, MP3's for your car or mobile phone (i.e. non-critical listening) then it would be best to use Repair Record mode.